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PHYSICS

Criticize Pure Reason in proximity of a Hollow-Core Fiber

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Pokémon forti, Pokémon deboli.
Sono distinzioni dettate dall'egoismo.
Gli allenatori davvero in gamba dovrebbero vincere coi loro preferiti.

Strong Pokémon, weak Pokémon.
That is only the selfish perception of people.
Truly skilled trainers should try to win with their favorites.

Karen, Johto Elite Four

Contents

Introduction

Your introduction text here.

Chapter 1

Theoretical Background

This chapter provides an overview of the theoretical background underlying the main topics addressed in this thesis.

The first part of the chapter focuses on conventional step-index optical fiber, with the aim of explaining the physics of mode confinement. Starting from the wave equation, the electromagnetic field behavior in the fibers is derived. This lead, under the appropriate approximations, to the description of linearly polarized LP modes.

The second part of the chapter is devoted to the theory of optical dipole trapping. The physical mechanism that allow optical fields to confine neutral atom is discussed, and expressions for the trapping potential are derived for both ideal two-level atoms and realistic multi-level atoms.

1.1 Optical Fibers

Waveguides are devices that employ dielectric materials with a high refractive index to confine and guide electromagnetic radiation, preventing the diffraction-induced spreading that occurs during free-space propagation. The electromagnetic wave that propagates inside a waveguide is called guided mode.

Optical fibers are cylindrically symmetric waveguides, and represent the most widespread class of optical waveguides. Following chapter III of Yariv et al. [1], this section aims to explain the physical principles underlying light confinement in optical fibers. Particular attention is devoted to the characterization of the guided modes in the weakly guiding approximation, which leads to the so-called linearly polarized (LP) modes.

1.1.1 Circularly symmetric waveguides

An optical fiber can be modeled as a cylindrical core of refractive index n_1 surrounded by a cylindrical cladding of refractive index n_2. The refractive index distribution is therefore

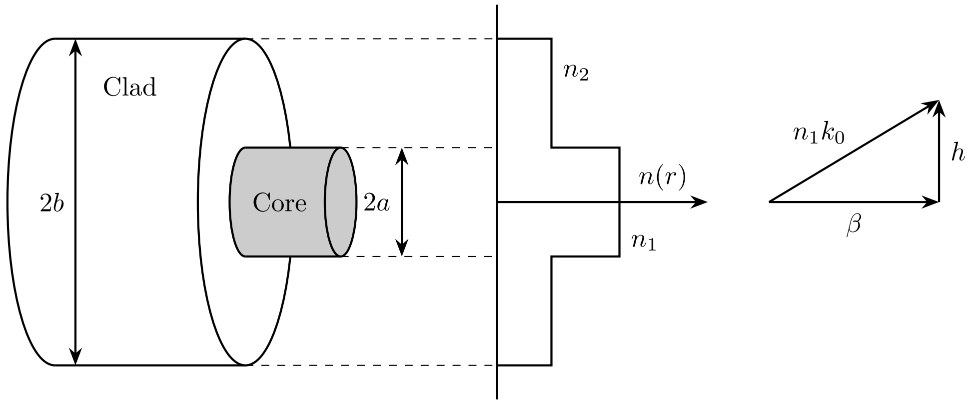


Figure 1.1: *Schematic representation of a step-index optical fiber (left), and wave vector diagram showing the geometric relationship between the wave number in the core* n_1k_0*, the longitudinal propagation constant* β*, and the transverse wave number* h *(right).*

described by the step-index profile

$$n(r) = \begin{cases} n_1, & 0 < r < a \\ n_2, & a < r < b, \end{cases} \tag{1.1}$$

where a is the radius of the core and b is the outer radius of the cladding. Usually the radius of the cladding is much larger than the radius of the core ($b \gg a$), thus the electromagnetic field can be considered zero at $r = b$. As an example, in a single-mode fiber guiding light with $\lambda = 1.5$ μm, the typical core radius is $a = 5$ μm while the cladding external radius is in the order of $b = 100$ μm.

1.1.1.1 The Cylindrical Wave Equation

Owing to the cylindrical symmetry of the system, it is convenient to work with the cylindrical coordinates:

$$\begin{cases} \mathbf{u_r} = & \hat{\mathbf{x}} \cos\phi + \hat{\mathbf{y}} \sin\phi \\ \mathbf{u}_\phi = & -\hat{\mathbf{x}} \sin\phi + \hat{\mathbf{y}} \cos\phi \\ \mathbf{u_z} = & \hat{\mathbf{z}}, \end{cases} \tag{1.2}$$

thus the electric and magnetic fields are expressed in terms of the components E_r, E_ϕ, E_z, H_r, H_ϕ and H_z. In cylindrical coordinates the unit vectors $\mathbf{u_r}$ and $\mathbf{u}_\phi$ are not constant vectors ($\partial\mathbf{u_r}/\partial\phi = \mathbf{u}_\phi$ and $\partial\mathbf{u}_\phi/\partial\phi = -\mathbf{u_r}$). This complicates the wave equation for the associated components of the field. Nevertheless, as $\mathbf{u_z} = \hat{\mathbf{z}}$ the equation for the z components remains simple. Assuming the dependence on the coordinate z and time t to be $e^{i(\omega t - \beta z)}$, namely:

$$\begin{bmatrix} \mathbf{E}(\mathbf{r}, t) \\ \mathbf{H}(\mathbf{r}, t) \end{bmatrix} = \begin{bmatrix} \mathbf{E}(r, \phi) \\ \mathbf{H}(r, \phi) \end{bmatrix} e^{i(\omega t - \beta z)}, \tag{1.3}$$

where β is referred to as the propagation constant of the mode, the wave equation for E_z and H_z is

$$(\nabla^2 + k^2)\begin{bmatrix} E_z \\ H_z \end{bmatrix} = 0, \tag{1.4}$$

where $k^2 = \omega^2 n^2/c^2$ and ∇^2 is the Laplacian operator in cylindrical coordinates. To solve the cylindrical wave equation, the standard approach is to solve equation (1.4) and then express E_r, E_ϕ, H_r and H_ϕ in terms of E_z and H_z through Maxwell's curl equations in cylindrical coordinates (Appendix A of [1]).

Using (1.3) and explicitly writing the transverse cylindrical Laplacian in (1.4), one gets:

$$\left(\frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial \phi^2} + (k^2 - \beta^2)\right)\begin{bmatrix} E_z \\ H_z \end{bmatrix} = 0. \tag{1.5}$$

Equation (1.5) is solved by assuming the separation of variables

$$F_z(r, \phi) = R(r)\Phi(\phi), \tag{1.6}$$

where $F_z = E_z, H_z$, obtaining

$$\Phi\frac{d^2R}{dr^2} + \frac{1}{r}\Phi\frac{dR}{dr} + \frac{1}{r^2}R\frac{d^2\Phi}{d\phi^2} + (k^2 - \beta^2)R(r)\Phi(\phi) = 0. \tag{1.7}$$

Dividing by $R\Phi$ and rearranging terms leads to

$$r^2\left(\frac{1}{R}\frac{d^2R}{dr^2} + \frac{1}{r}\frac{1}{R}\frac{dR}{dr} + (k^2 - \beta^2)\right) = -\frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2}. \tag{1.8}$$

The last equation is in the form $f(r) = g(\phi)$. Therefore, since the left-hand side depends only on r and the right-hand side only on ϕ, both sides must be equal to the same constant:

$$\begin{cases} r^2\left(\frac{1}{R}\frac{d^2R}{dr^2} + \frac{1}{r}\frac{1}{R}\frac{dR}{dr} + (k^2 - \beta^2)\right) = l^2 \\ -\frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2} = l^2 \end{cases}. \tag{1.9}$$

The angular part has a simple exponential solution:

$$\Phi(\phi) = C_1 e^{+il\phi} + C_2 e^{-il\phi}, \tag{1.10}$$

where C_1 and C_2 are constants that are determined by the boundary conditions of the problem. To be physically meaningful, $\Phi(\phi)$ must be single valued ($\Phi(\phi) = \Phi(\phi + 2\pi)$), thus the value of l is constrained to

$$l = 0, 1, 2, ... \tag{1.11}$$

A set of elementary solutions can therefore be expressed in the form

$$\begin{bmatrix} E_z \\ H_z \end{bmatrix} = R(r)e^{\pm il\phi},$$

where the differential equation that describes the radial part is

$$\frac{d^2R}{dr^2} + \frac{1}{r}\frac{dR}{dr} + \left(k^2 - \beta^2 - \frac{l^2}{r^2}\right) R = 0, \tag{1.12}$$

which is recognized as Bessel's differential equation, and its solutions are a linear combination of Bessel functions of order l.

1.1.1.2 The Bessel functions

The radial equation (1.12) admits different classes of solutions depending on the sign of the quantity $k^2 - \beta^2$:

- for $k^2 - \beta^2 > 0$, the solutions are given by the ordinary Bessel functions of the first and second kind of order l:

$$R(r) = \bar{C}_1 J_l(hr) + \bar{C}_2 Y_l(hr); \tag{1.13}$$

- for $k^2 - \beta^2 < 0$, the solutions are given by the modified Bessel functions of the first and second kind, of order l:

$$R(r) = \bar{C}_1 I_l(qr) + \bar{C}_2 K_l(qr), \tag{1.14}$$

where $h^2 = k^2 - \beta^2$ and $q^2 = \beta^2 - k^2$, and $\bar{C}_1$ and $\bar{C}_2$ are arbitrary constants determined by the boundary conditions.

The behavior of the ordinary and modified Bessel functions is illustrated in Figure 1.2. To proceed with the physical solution, it is useful to consider their asymptotic forms for both large and small arguments.

For $x \gg 1, l$, the asymptotic expressions are

$$\begin{aligned}
J_l(x) &\to \left(\frac{2}{\pi x}\right)^{1/2} \cos\left(x - \frac{l\pi}{2} - \frac{\pi}{4}\right) \\
Y_l(x) &\to \left(\frac{2}{\pi x}\right)^{1/2} \sin\left(x - \frac{l\pi}{2} - \frac{\pi}{4}\right) \\
I_l(x) &\to \left(\frac{1}{2\pi x}\right)^{1/2} e^{x} \\
K_l(x) &\to \left(\frac{\pi}{2x}\right)^{1/2} e^{-x}
\end{aligned} \tag{1.15}$$

For $x \ll 1$, the corresponding approximations are

$$
\begin{aligned}
J_l(x) &\to \frac{1}{l!}\left(\frac{x}{2}\right)^l \\
Y_0(x) &\to \frac{2}{\pi}\left(\ln\frac{x}{2} + 0.5772\ldots\right) \\
Y_l(x) &\to -\frac{(l-1)!}{\pi}\left(\frac{2}{x}\right)^l \qquad l = 1, 2, 3, \ldots \\
I_l(x) &\to \frac{1}{l!}\left(\frac{x}{2}\right)^l \\
K_0(x) &\to -\left(\ln\frac{x}{2} + 0.5772\ldots\right) \\
K_l(x) &\to \frac{(l-1)!}{2}\left(\frac{2}{x}\right)^l
\end{aligned}
\tag{1.16}
$$

These asymptotic behaviors play a crucial role in the construction of guided-mode solutions. In particular, the functions $Y_l(x)$ and $K_l(x)$ diverge as $x \to 0$, whereas $K_l(x)$ decays exponentially for large arguments. Consequently, only specific combinations of Bessel functions satisfy the physical requirements for confined modes.

Another important property is that the derivative of a Bessel function of order l can be expressed as a combination of Bessel functions of orders $l-1$ and $l+1$. Since the transverse field components E_r, E_ϕ, H_r, and H_ϕ are obtained from spatial derivatives of E_z and H_z, they are generally described by combinations of Bessel functions of different orders.

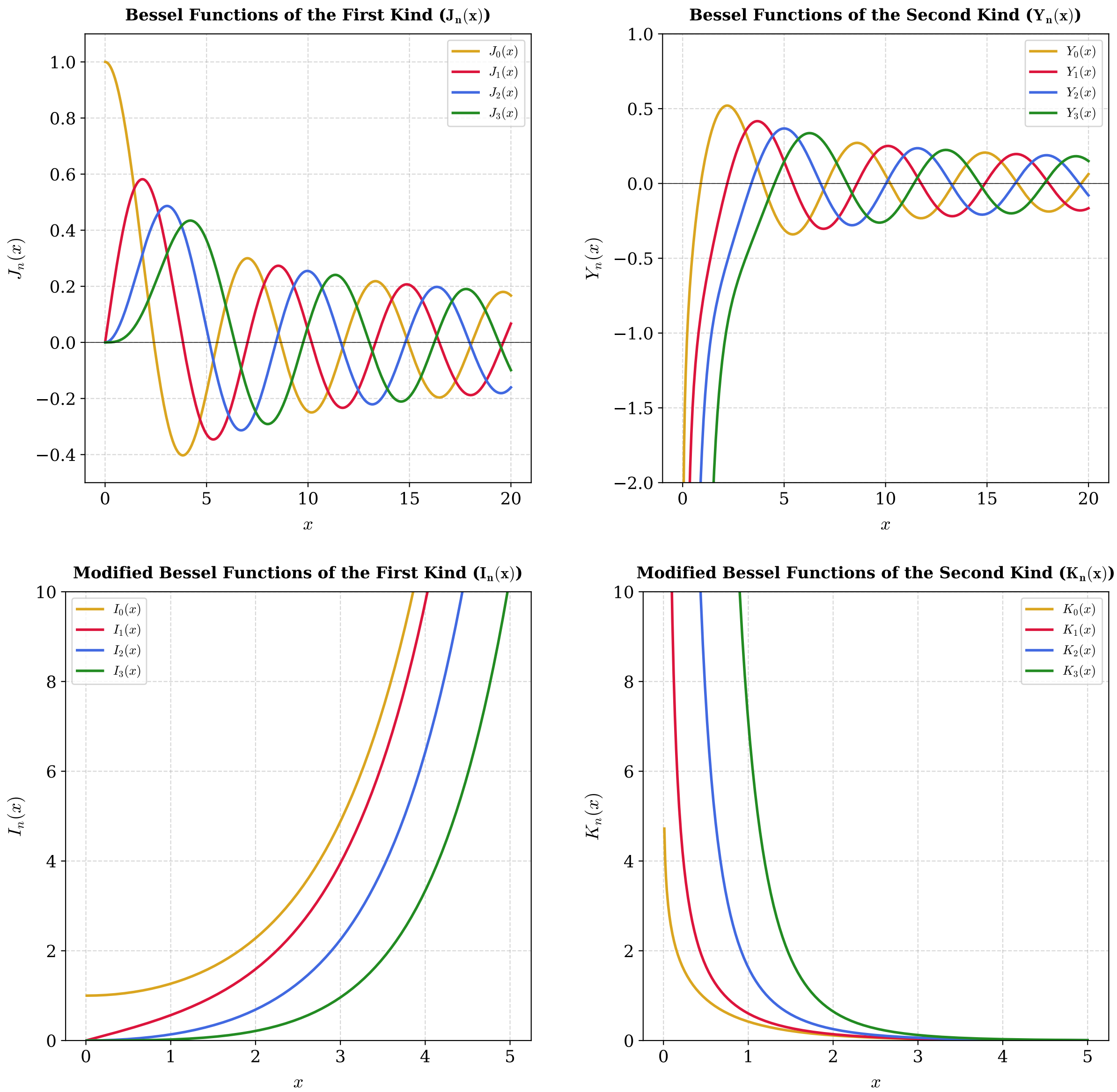


Figure 1.2: *Behavior of ordinary and modified Bessel functions for the first four integer orders, $n \in \{0, 1, 2, 3\}$. Top panels: Ordinary Bessel functions of the first kind, $J_n(x)$ (left), and second kind, $Y_n(x)$ (right). Bottom panels: Modified Bessel functions of the first kind, $I_n(x)$ (left), and second kind, $K_n(x)$ (right).*

1.1.2 The Weakly Guiding Approximation

The solution of the complete problem of electromagnetic guided modes in circular waveguides is cumbersome, and it leads to the so-called hybrid modes indicated by EH and HE, which are a combination of modes with the electric field transverse to the propagation of the wave (TE) and modes with the magnetic field transverse to the propagation direction (TM).

However, for a large variety of fibers, the core refractive index is only slightly higher than the refractive index of the cladding:

$$\frac{n_1 - n_2}{n_1} \ll 1, \tag{1.17}$$

a condition commonly referred to as weakly guiding approximation. Under this assumption, the exact modes can be accurately approximated by a simpler set of modes, known as linearly polarized (LP) modes, first introduced in the work of Gloge [2].

Using the relation $k^2 = (n(r)\,k_0)^2$ where $k_0 = \omega/c$, equation (1.5) can be rewritten as

$$[\nabla_t^2 + ((n(r)\,k_0)^2 - \beta^2)] \begin{bmatrix} E_z \\ H_z \end{bmatrix} = 0. \tag{1.18}$$

Looking for a guided mode in the fiber is equivalent to seeking a solution that exhibits an oscillatory behavior in the core while exhibiting evanescent decay in the cladding. From the properties of second-order differential equations, these requirements imply:

$$\begin{aligned} r > a \quad &\text{vanishing} \quad \Longrightarrow (n_2 k_0)^2 - \beta^2 < 0, \\ r < a \quad &\text{oscillating} \quad \Longrightarrow (n_1 k_0)^2 - \beta^2 > 0. \end{aligned} \tag{1.19}$$

Consequently, the propagation constant of a guided mode must satisfy

$$n_2 k_0 < \beta < n_1 k_0, \tag{1.20}$$

which in weakly guiding regime becomes

$$n_2 k_0 \simeq \beta \simeq n_1 k_0. \tag{1.21}$$

At this point, it is convenient to introduce the transverse wave numbers in the core (h) and in the cladding (q):

$$h = \sqrt{n_1^2\,k_0^2 - \beta^2}, \tag{1.22}$$

$$q = \sqrt{\beta^2 - n_2^2\,k_0^2}. \tag{1.23}$$

The parameter h describes the transverse oscillation of the field within the core (see Figure 1.1), whereas q characterizes the decay of the mode in the cladding.
Equation (1.21) implies that

$$h, q \ll \beta. \tag{1.24}$$

Physically this means that the rays are approximately parallel to the fiber axis. Therefore, the longitudinal components of the electric and magnetic field are far weaker than the transverse ones, and the guided modes are approximately transverse electromagnetic (TEM) [3].

The objective is now to identify a complete set of approximate modal solutions in the weakly guiding limit. The approach is to focus on solutions where either the x or the y component of the field vanishes, thus looking for x-polarized or y-polarized solutions. The approximation made by Gloge [2] is based on postulating for the transverse component a simple dependence on a single Bessel function. Matching the requirements (1.19) with the Bessel equation solution condition (1.13) and (1.14), one deduces that the mode must be described by ordinary Bessel functions inside the core and by modified Bessel functions in the cladding. Furthermore, physically acceptable solutions must remain finite at the fiber axis and vanish at large radial distances. By comparison with the asymptotic behavior in (1.15) and (1.16), only J_l is admissible in the core region, whereas only K_l satisfies the required decay condition in the cladding.

For a mode polarized along the (y)-direction, the transverse electric field is therefore postulated as

$$\begin{aligned} E_x &= 0 \\ E_y &= \begin{cases} A J_l(hr) e^{\pm il\phi} e^{i(\omega t - \beta z)}, & r < a \\ B K_l(qr) e^{\pm il\phi} e^{i(\omega t - \beta z)}, & r > a \end{cases} , \end{aligned} \tag{1.25}$$

where A and B are constants that are determined from the continuity condition at the core–cladding interface.
It is worth noting that, for each value of l, equation (1.25) describes two independent azimuthal solutions associated to $e^{+il\phi}$ and $e^{-il\phi}$. Taking into account the two possible independent linear polarizations, a total of four independent basis functions are associated with each value of l.

The remaining electromagnetic field components can be obtained from Maxwell's equations. In particular, the magnetic field is given by [1]:

$$\begin{cases} H_x = \frac{-i}{\omega\mu} \frac{\partial}{\partial z} E_y = -\frac{\beta}{\omega\mu} E_y \\ H_y \simeq 0 \\ H_z = \frac{i}{\omega\mu} \frac{\partial}{\partial x} E_y \end{cases} , \tag{1.26}$$

and similarly, the longitudinal electric field component can be expressed as

$$E_z = \frac{i}{\omega\epsilon}\frac{\partial}{\partial y}H_x = \frac{-i\beta}{\omega^2\mu\epsilon}\frac{\partial}{\partial y}E_y. \tag{1.27}$$

Solving the differential equations for E_z and H_z yields the complete set of basis fields in the weakly guiding approximation [1].

In the core ($r < a$):

$$\begin{cases} E_x = 0 \\ E_y = AJ_l(hr)e^{\pm il\phi}e^{i(\omega t-\beta z)} \\ E_z = \pm\frac{h}{\beta}\frac{A}{2}[J_{l+1}(hr)e^{\pm i(l+1)\phi} + J_{l-1}(hr)e^{\pm i(l-1)\phi}]e^{i(\omega t-\beta z)} \end{cases}, \tag{1.28}$$

$$\begin{cases} H_x = -\frac{\beta}{\omega\mu}AJ_l(hr)e^{\pm il\phi}e^{i(\omega t-\beta z)} \\ H_y = 0 \\ H_z = -\frac{ih}{\omega\mu}\frac{A}{2}[J_{l+1}(hr)e^{\pm i(l+1)\phi} - J_{l-1}(hr)e^{\pm i(l-1)\phi}]e^{i(\omega t-\beta z)} \end{cases}. \tag{1.29}$$

In the cladding ($r > a$):

$$\begin{cases} E_x = 0 \\ E_y = BK_l(qr)e^{\pm il\phi}e^{i(\omega t-\beta z)} \\ E_z = \frac{q}{\beta}\frac{B}{2}[K_{l+1}(qr)e^{\pm i(l+1)\phi} - K_{l-1}(qr)e^{\pm i(l-1)\phi}]e^{i(\omega t-\beta z)} \end{cases}, \tag{1.30}$$

$$\begin{cases} H_x = -\frac{\beta}{\omega\mu}BK_l(qr)e^{\pm il\phi}e^{i(\omega t-\beta z)} \\ H_y = 0 \\ H_z = -\frac{iq}{\omega\mu}\frac{B}{2}[K_{l+1}(qr)e^{\pm i(l+1)\phi} + K_{l-1}(qr)e^{\pm i(l-1)\phi}]e^{i(\omega t-\beta z)} \end{cases}. \tag{1.31}$$

Considering that $E_z/E_y \propto h/\beta$, $H_z/H_x \propto h/\beta$ in the core and $E_z/E_y \propto q/\beta$, $H_z/H_x \propto q/\beta$ in the cladding, since $h, q \ll \beta$, E_y and H_x are the dominant components, and the mode is approximately TEM, consistently with the considerations given above.

Considering that $E_\phi = -E_x\sin(\phi) + E_y\cos(\phi)$, for a y-polarized mode $E_\phi \propto E_y$, thus the radial continuity condition at $r = a$ reduces to the continuity condition of E_y, which leads to

$$B = A\frac{J_l(ha)}{K_l(qa)}. \tag{1.32}$$

An identical condition is obtained from the continuity of H_ϕ.

On the other hand, the continuity condition of E_z at $r = a$ must hold for every value of the azimuthal coordinate ϕ. Consequently, the coefficients of $\exp[\phi(l+1)\phi]$ and $\exp[\pm i(l-1)\phi]$

must be matched separately. This leads to

$$ha\frac{J_{l+1}(ha)}{J_l(ha)} = qa\frac{K_{l+1}(qa)}{K_l(qa)}, \tag{1.33}$$

$$ha\frac{J_{l-1}(ha)}{J_l(ha)} = -qa\frac{K_{l-1}(qa)}{K_l(qa)}, \tag{1.34}$$

which are equivalent as a consequence of the recurrence relations satisfied by the Bessel functions [1]. The same characteristic equations are obtained by imposing continuity of H_z.

Finally, repeating the same procedure for an x-polarized mode leads to identical conditions, demonstrating that both polarizations share the same modal eigenvalues.

1.1.3 The LP modes

Equation (1.33) depends only on β through the transverse wave numbers h and q. It yields an infinite set of discrete solutions for each value of l. Therefore, the propagation constants are labeled as:

$$\beta_{lm} \quad : \quad l = 0, 1, 2, ... \quad m = 1, 2, 3, ... \tag{1.35}$$

Their corresponding modes are called Linearly Polarized modes, denoted by:

$$LP_{lm} \quad : \quad l = 0, 1, 2, ... \quad m = 1, 2, 3, ... \tag{1.36}$$

Since the condition in (1.33) holds for both x-polarized and y-polarized waves, the modes LP_{lm-x} and LP_{lm-y} are degenerate sharing the same propagation constant β_{lm}. This degeneracy is a direct consequence of the circular symmetry of the fiber. Furthermore, accounting for the degeneracy associated with the azimuthal phase term $e^{\pm il\phi}$, the propagation constant β_{lm} and the mode LP_{lm} are four-fold degenerate. A notable exception occurs for the fundamental case $l = 0$, for which the exponential is equal to unity independently of the sign of $l\phi$, reducing β_{0m} and LP_{0m} to a two-fold degeneracy.

1.1.3.1 Characteristic equation of the LP modes

In order to determine the allowed values of β_{lm}, the characteristic equation (1.33) must be solved. Since the transverse wave numbers h and q are interdependent, it is convenient to express them in terms of a dimensionless parameter. For this purpose, the normalized frequency, also called fiber V parameter, is introduced and defined as:

$$V = k_0 a\sqrt{n_1^2 - n_2^2} = \frac{2\pi a}{\lambda}\sqrt{n_1^2 - n_2^2} = \sqrt{(ha)^2 + (qa)^2}. \tag{1.37}$$

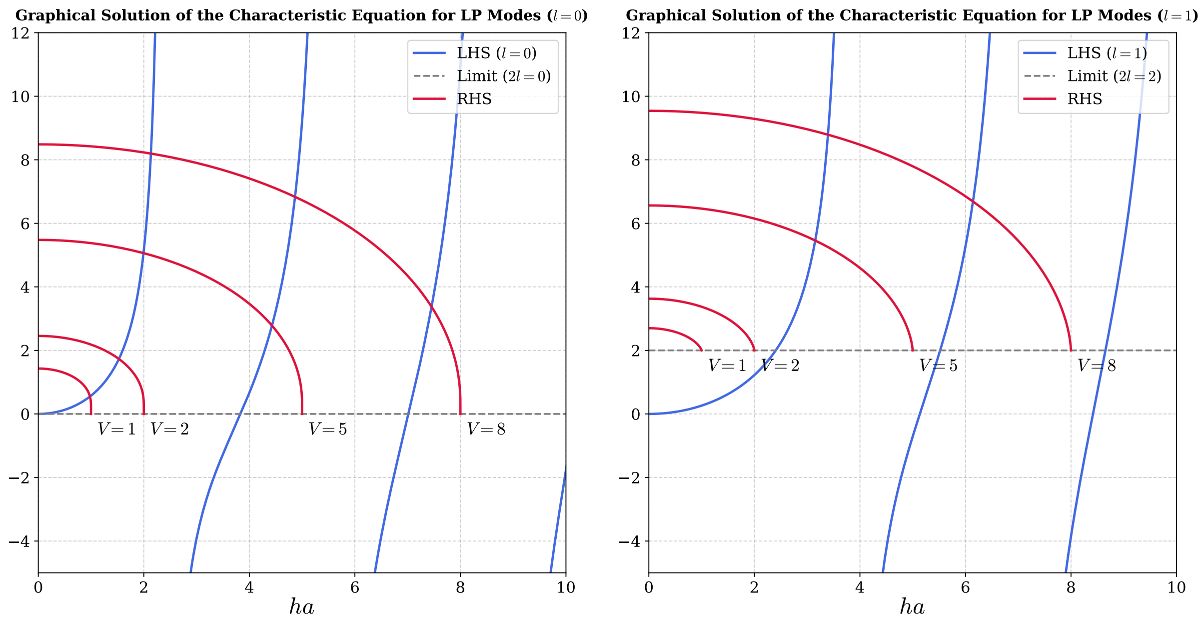


Figure 1.3: *Graphical solution of the characteristic equation* (1.33) *of the LP modes, for* $l = 0$ *(left) and* $l = 1$ *(right). The blue curves represent the left-hand side (LHS) of equation* (1.33) *for the given* l. *The red curves represent the right-hand side (RHS) for* $V = 2, 5, 8$. *The gray dashed line indicates the limit* $qa \to 0$ *of the LHS.*

The V parameter is an intrinsic property of the fiber determined by its structural and material characteristics. As shown below, it governs the number of guided modes and determines their respective confinement factor.

Using the V parameter, the cladding variable qa can be expressed as:

$$qa = \sqrt{V^2 - (ha)^2}, \tag{1.38}$$

and substituting it back in equation (1.33), the characteristic equation is reduced to a function of the single variable ha, allowing for a graphical or numerical solution. Moreover, equation (1.38) explicitly restricts the domain of the left-hand side of (1.33) to $ha < V$.

Figure 1.3 shows the graphical solution for $l = 0$ (left) and $l = 1$ (right). For $l = 0$, the left-hand side of equation (1.33) vanishes at the roots of $J_1(ha)$, located at $ha = 0, 3.832, 7.016 \ldots$, whereas the denominator vanishes at the roots of $J_0(ha)$ which occur at $ha = 2.405, 5.520, 8.654 \ldots$ corresponding to vertical asymptotes. Therefore, in the interval $ha \in [0, 2.405]$ the left-hand side is a strictly increasing positive function going from 0 to $+\infty$. Conversely, the right-hand side, at $ha = 0$ assumes the positive value $VK_1(V)/K_0(V)$ and monotonically decreases to 0 at $ha = V$. Consequently, an intersection point always exists, even for very small values of V, implying that the fundamental LP_{01} mode is always guided by an optical fiber regardless of its V parameter.

Using (1.16), one derives that for $l \geq 1$, the right-hand side asymptotically approaches $2l$ as $ha \to V$. Under these conditions, an intersection is no longer guaranteed for every value of V. In particular for $V < 2.405$ the only possible guided mode is the LP_{01}. As V increases beyond 2.405, also the LP_{11} appears, therefore $V = 2.405$ is called the V cutoff value of the mode LP_{11}.

The cutoff value V_{cutoff} for a given mode can be determined by considering the condition at which the field is no longer evanescent in the cladding, corresponding to $q = 0$. Since $V = ha$ when $q = 0$, the cutoff condition can be obtained by imposing $q = 0$ in equation (1.34), yielding

$$V_{\text{cutoff}} \frac{J_{l-1}(V_{\text{cutoff}})}{J_l(V_{\text{cutoff}})} = 0. \tag{1.39}$$

For $l \geq 1$, in the limit $V_{\text{cutoff}} \to 0$ (see (1.16)) the left-hand side evaluates to $2l$ therefore for $l \geq 1$ the cutoff is strictly larger than zero. For $l = 0$ using the property $J_{-1}(V) = -J_1(V)$, for $V_{\text{cutoff}} \to 0$ one gets $0 = 0$, confirming that the mode LP_{01} is guided independently of the value of V.

Excluding the special case of the LP_{01}, equation (1.39) leads to

$$J_{l-1}(V_{\text{cutoff}}) = 0. \tag{1.40}$$

Therefore, the cutoff value of the LP_{lm} mode is given by the m-th zero of the Bessel function J_{l-1} if $l \neq 0$, and by the $(m-1)$-th zero of J_{l-1} if $l = 0$, while the LP_{01} mode is always guided. Here, the zero at the origin is not included in the enumeration of the zeros of the Bessel functions. The cutoff values of several low-order LP modes are reported in Table 1.1.

It is worth noting that equation (1.34) is employed to obtain the cutoff condition rather than (1.33) because its right-hand side conveniently vanished as $ha \to V$, making the cutoff directly associated to the roots of ordinary Bessel functions of the first kind.

Table 1.1: *Cutoff values of V for low-order LP modes.*

V_{cutoff}	$m = 1$	$m = 2$	$m = 3$	$m = 4$
$l = 0$	0	3.832	7.016	10.173
$l = 1$	2.405	5.520	8.654	11.792
$l = 2$	3.832	7.016	10.173	13.323
$l = 3$	5.136	8.417	11.620	14.796
$l = 4$	6.379	9.760	13.017	16.224

1.1.3.2 Power Flow of the LP modes

Prior to evaluating the power flow of the LP modes, a clarification regarding the basis chosen for the spatial representation of the field is necessary. So far, following the formalism in [1], a basis in which the azimuthal dependence is expressed by the exponential $e^{\pm il\phi}$, for the two degenerate solution $\pm l$ has been utilized. This representation is convenient for calculation, however, it leads to a basis in which the modes have an azimuthal phase dependence, or equivalently modes that carry a defined orbital angular momentum (OAM) $l\hbar$ per photon [4]. In typical experimental conditions, however, optical fibers are coupled with standard Gaussian beams, which carry zero macroscopic OAM due to their lack of azimuthal phase gradients. Therefore, to conserve the net angular momentum at the fiber input, the coupling excites the two azimuthal counter-propagating spatial modes $+l$ and $-l$ with identical amplitudes. Consequently, a basis employing trigonometric functions ($\cos(l\phi)$ and $\sin(l\phi)$) is physically more representative of typical intensity profiles and is widely adopted in the literature. These functions represent a linear superposition of counter-propagating azimuthal modes carrying opposite OAM, yielding a net-zero OAM state:

$$\begin{aligned} \cos(l\phi) &= \frac{e^{il\phi} + e^{-il\phi}}{2}, \\ \sin(l\phi) &= \frac{e^{il\phi} - e^{-il\phi}}{2i}. \end{aligned} \tag{1.41}$$

Therefore, the trigonometric basis is adopted for the remainder of this analysis. While a pure $e^{\pm il\phi}$ basis implies a ring-shaped intensity profile devoid of azimuthal modulation, the trigonometric basis captures the interference between opposite OAM states, producing the characteristic $2l$-lobe intensity patterns routinely observed experimentally.

The power flow of the LP modes is evaluated via the longitudinal component of the time-averaged Poynting vector:

$$S_z = \frac{1}{2}\,\mathrm{Re}[E_x H_y^* - E_y H_x^*]. \tag{1.42}$$

Using the field expressions (1.28), (1.29), (1.30) and (1.31), the trigonometric azimuthal basis, the power flux density becomes:

$$S_z^{(lm)} = \frac{\beta}{2\omega\mu} \begin{cases} |A|^2 J_l^2(hr)\cos^2(l\phi) & r < a \\ \\ |B|^2 K_l^2(qr)\cos^2(l\phi) & r > a \end{cases} , \tag{1.43}$$

where the dependence on m is implicitly contained in h and q. The power guided in the core and cladding is obtained by integrating the Poynting vector over the respective cross-sections:

$$P_{\text{core}} = \int_0^{2\pi}\int_0^a S_z r\,dr\,d\phi, \tag{1.44}$$

$$P_{\text{clad}} = \int_0^{2\pi}\int_a^\infty S_z r\,dr\,d\phi. \tag{1.45}$$

Evaluating these integrals [1], and substituting B from (1.32), the result is:

$$P_{\text{core}} = \frac{\beta}{2\omega\mu}\pi a^2 |A|^2 \left[J_l^2(ha) - J_{l-1}(ha)J_{l+1}(ha)\right], \tag{1.46}$$

$$P_{\mathrm{clad}} = \frac{\beta}{2\omega\mu}\pi a^2 |A|^2 \left[-J_l^2(ha) - \left(\frac{h}{q}\right)^2 J_{l-1}(ha) J_{l+1}(ha) \right]. \tag{1.47}$$

Summing these contributions gives the total power flow:

$$P_{\mathrm{tot}} = \frac{\beta}{2\omega\mu}\pi a^2 |A|^2 \left(1 + \frac{h^2}{q^2}\right) \left[-J_{l-1}(ha) J_{l+1}(ha) \right]. \tag{1.48}$$

By multiplying equations (1.33) and (1.34) side by side, one gets:

$$(ha)^2 \frac{J_{l+1}(ha) J_{l-1}(ha)}{J_l^2(ha)} = -(qa)^2 \frac{K_{l+1}(qa) K_{l-1}(qa)}{K_l^2(qa)}. \tag{1.49}$$

Given that the modified Bessel functions K_l are always positive (see Figure 1.2), the quantity $J_{l-1}(ha)J_{l+1}(ha)$ is always negative for the allowed values of ha, therefore the total power flow (1.48) is always positive as physically expected.
The core confinement factor is defined as $\Gamma_1 = P_{\mathrm{core}}/P_{\mathrm{tot}}$

$$\Gamma_1 = \frac{P_{\mathrm{core}}}{P_{\mathrm{tot}}} = \frac{1}{V^2} \left((qa)^2 - \frac{(qa)^2 J_l^2(ha)}{J_{l-1}(ha) J_{l+1}(ha)} \right). \tag{1.50}$$

Γ_1 gives a measure of how much of the mode is travelling in the core rather than in the cladding. This parameter is important for calculating the dispersion and attenuation properties of the fiber, which are critical parameters in the design of fibers for telecommunication systems.
The dependence of Γ_1 on V is not trivial as both h and q depend on V through the solution of the characteristic equation (1.33). Generally, the confinement factor Γ_1 increases monotonically with V. The confinement factor at cutoff is obtained by taking the limit $q \to 0$ in (1.50), obtaining:

$$\Gamma_1 = \begin{cases} 0 & \text{for} \quad l = 0, 1 \\ 1 - \frac{1}{l} & \text{for} \quad l > 1 \end{cases}. \tag{1.51}$$

For example the mode LP_{21} at cutoff has a confinement efficiency of 0.5. Figure 1.4 illustrates the behavior of the confinement factor as a function of the V parameter for some low-order LP modes.

Finally, the calculated intensity profiles for selected LP modes are mapped in Figure 1.5. The figure highlights how the azimuthal index l governs the number of diametric nodal lines crossing the beam axis, whereas the radial index m dictates the number of concentric local intensity maxima along the radial coordinate.

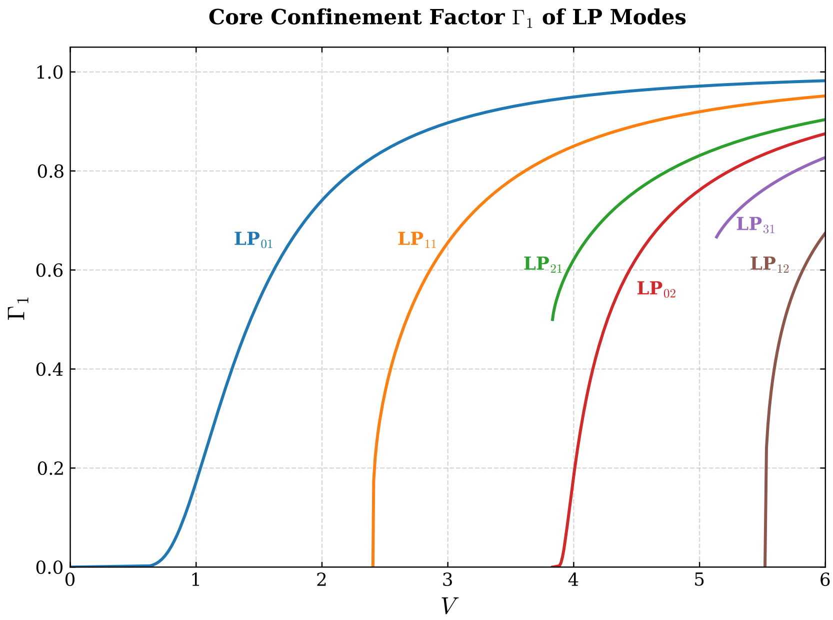


Figure 1.4: *Confinement factor* Γ_1 *of LP modes as a function of the fiber* V *parameter. Note that for the modes with* $l > 1$ *(*LP_{21}*,* LP_{31}*) the confinement at cutoff is different from* 0.

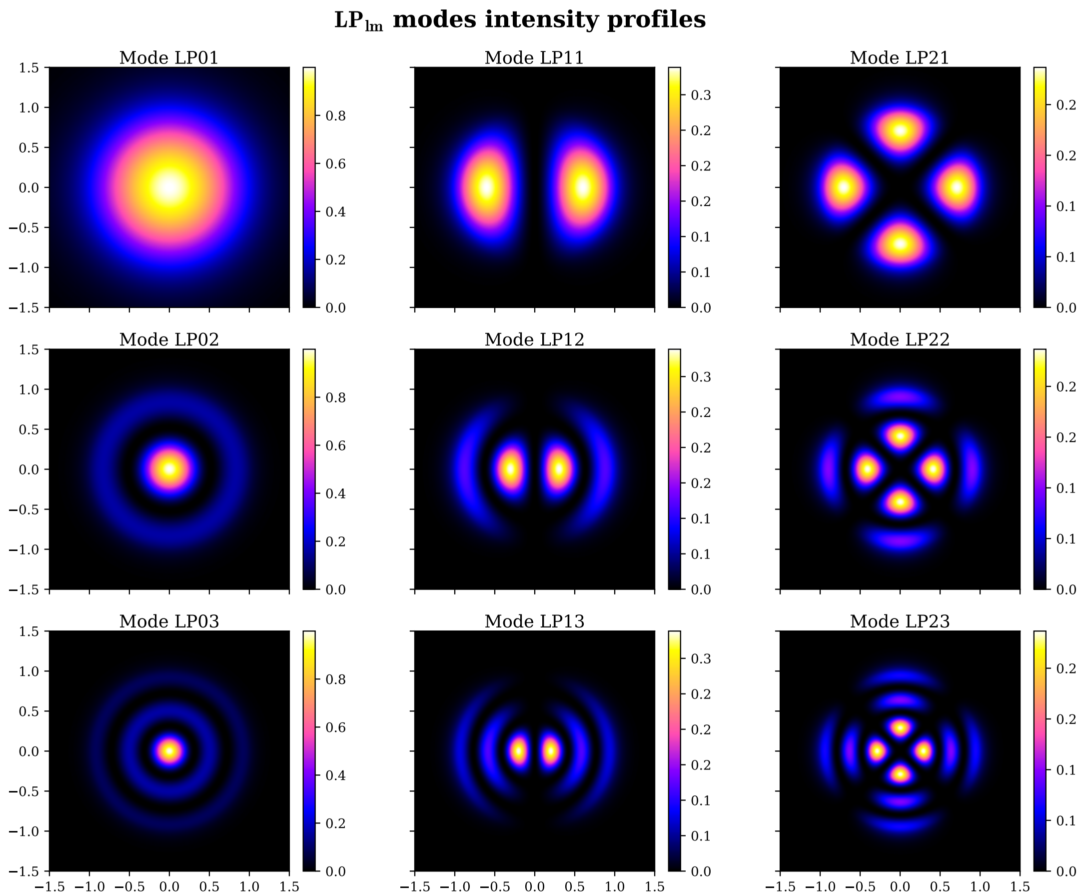


Figure 1.5: *Intensity profiles of the LP modes with* $l = 0, 1, 2$ *and* $m = 1, 2, 3$*. Each mode* LP_{lm} *is calculated with* $V^{(lm)} = V^{(lm)}_{cutoff} + 2$*. The* x, y *coordinates are expressed in units of fiber core radius.*

1.2 Optical Dipole Trap

Optical traps are experimental tools that allow the confinement of atom or ions within a localized region of space, under the condition that these particles have a kinetic energy lower than the trap potential depth (typically below 1mK). For the work presented in this thesis, among all possible trapping techniques, the focus will be strictly placed on Optical Dipole Traps (ODTs) for neutral atoms.

Optical dipole traps rely on the electric dipole interaction of the atoms with far-detuned light. Following the work of Grimm et al. [5], this section reviews the theory behind the working mechanism of this technique, particularly in the case of red-detuned ODTs.

1.2.1 Oscillator model

The optical dipole force arises from the interaction of the induced atomic momentum with the intensity gradient of the light field. It is a conservative force, thus in the following its potential energy will be derived modelling the atom as an oscillator subjected to a classical oscillating electric field.

1.2.1.1 Interaction between induced dipole moment and driving field

The laser field is modeled as the oscillating electric field:

$$\mathbf{E}(\mathbf{r},t) = \hat{\mathbf{e}}\mathcal{E}(\mathbf{r})e^{-i\omega t} + c.c. \ , \tag{1.52}$$

where $\hat{\mathbf{e}}$ is the unitary polarization vector and $\mathcal{E}(r)$ is the complex amplitude of the field. Denoting the atomic polarizability as α, the induced dipole momentum of the atom is given by:

$$\mathbf{p}(\mathbf{r},t) = \alpha\mathbf{E}(\mathbf{r},t) = \hat{\mathbf{e}}\alpha\mathcal{E}(\mathbf{r})e^{-i\omega t} + c.c. \tag{1.53}$$

In the semiclassical framework, the interaction of a neutral atom with a laser light field can be described using the dipole approximation. The interaction Hamiltonian takes the form $H_{\text{dip}} = -\mathbf{p}\cdot\mathbf{E}$. For an induced dipole, where the dipole moment depends linearly on the external field, the potential energy is obtained by integrating the work required to induce the dipole as the electric field is increased from zero to its final value:

$$U_{\text{dip}}(t) = \int_0^{\mathbf{E}} -\alpha\mathbf{E}'\cdot d\mathbf{E}' = -\frac{1}{2}\alpha E^2 = -\frac{1}{2}\mathbf{p}\cdot\mathbf{E}\,. \tag{1.54}$$

Averaging over a field period [5]:

$$U_{\rm dip} = -\frac{1}{2}\langle \mathbf{p}\cdot\mathbf{E}\rangle = -\frac{1}{2\epsilon_0 c}\mathrm{Re}[\alpha]I, \tag{1.55}$$

where $I = 2\epsilon_0 c|\mathcal{E}|^2$ is the time-averaged field intensity.
This result shows that an atom in an oscillating laser field experiences a force proportional to the field intensity I, and to the real part of the polarizability, which represent the in-phase component of the dipole oscillation, responsible for the dispersive properties of the interaction.

It follows that the dipole force is a conservative force proportional to the electric field intensity gradient:

$$\mathbf{F}_{\rm dip}(\mathbf{r}) = -\nabla U_{\rm dip} = \frac{1}{2\epsilon_0 c}\mathrm{Re}[\alpha]\nabla I(\mathbf{r}). \tag{1.56}$$

The dipole oscillation absorbs a power

$$P_{\rm abs} = \langle \dot{\mathbf{p}}\mathbf{E}\rangle = 2\omega \mathrm{Im}[\alpha\mathcal{E}^2] = \frac{\omega}{2\epsilon_0 c}\mathrm{Im}[\alpha]I, \tag{1.57}$$

which is proportional to the out-of-phase component of the oscillating dipole. The absorbed power can be seen as series of absorption and spontaneous emission of photons with energy $\hbar\omega$. Therefore, the total scattering rate is given by

$$\Gamma_{\rm sc}(\mathbf{r}) = \frac{P_{\rm abs}}{\hbar\omega} = \frac{1}{\hbar\epsilon_0 c}\mathrm{Im}[\alpha]I(\mathbf{r}). \tag{1.58}$$

1.2.1.2 Atomic polarizability

The polarizability α can be calculated by considering the classical Lorentz oscillator model for the atom, according to which the polarizability can be calculated by solving the differential equation

$$\ddot{x} + \Gamma(\omega)\dot{x} + \omega_0^2 x = -\frac{eE(t)}{m_e}, \tag{1.59}$$

Using $E(t) = \mathcal{E}_0\exp(-i\omega t)$, thus looking for a solution in the form $x(t) = x_0\exp(-i\omega t)$, equation (1.59) gives

$$x_0 = -\frac{e\mathcal{E}_0}{m_e}\frac{1}{\omega_0^2 - \omega^2 - i\omega\Gamma(\omega)}. \tag{1.60}$$

Using $p = -ex(t) = -ex_0\exp(-i\omega t)$ and (1.60), one can easily obtain the polarizability

$$\alpha(\omega) = \frac{e^2}{m_e}\frac{1}{\omega_0^2 - \omega^2 - i\omega\Gamma(\omega)}, \tag{1.61}$$

where $\Gamma(\omega)$ is the classical damping rate due to the radiative energy loss:

$$\Gamma^{\text{class.}}(\omega) = \frac{e^2\omega^2}{6\pi\epsilon_0 m_e c^3}. \tag{1.62}$$

Defining the on-resonance spontaneous emission rate $\Gamma \equiv \Gamma^{\text{class.}}(\omega_0) = (\omega_0/\omega)^2\,\Gamma^{\text{class.}}(\omega)$, the polarizability becomes

$$\alpha(\omega) = 6\pi\epsilon_0 c^3 \frac{\Gamma/\omega_0^2}{\omega_0^2 - \omega^2 - i(\omega^3/\omega_0^2)\Gamma}. \tag{1.63}$$

When considering a full quantum picture of a two level system subjected to classical radiation, Γ should be substituted with the spontaneous emission rate

$$\Gamma = \frac{\omega_0^3}{3\pi\epsilon_0\hbar c^3}|\langle e|\,\hat{d}\,|g\rangle|^2, \tag{1.64}$$

where $\hat{d}$ is the dipole moment operator.
Nevertheless, optical dipole traps work in far-detuned low saturation regime. Consequently, the transitions to the excited state are strongly suppressed, and the atom can be considered as occupying the ground state only. In this limit, the atomic equation of motion reduces to a classical damped harmonic oscillator model described by equation 1.59 (for a rigorous derivation see Chapter V of Steck [6]). Therefore, the classical polarizability given by equation 1.63 represent a well suited approximation for ODTs.

1.2.1.3 Dipole potential and scattering rate

Substituting α form (1.63) in (1.55) and (1.58) one gets the expressions of the dipole potential and the scattering rate of a dipole trap as explicit function of ω, ω_0 and Γ:

$$U_{\text{dip}}(\mathbf{r}) = -\frac{3\pi c^2}{2\omega_0^3}\left(\frac{\Gamma}{\omega_0 - \omega} + \frac{\Gamma}{\omega_0 + \omega}\right) I(\mathbf{r}), \tag{1.65}$$

$$\Gamma_{\text{sc}}(\mathbf{r}) = \frac{3\pi c^2}{2\hbar\omega_0^3}\left(\frac{\omega}{\omega_0}\right)^3\left(\frac{\Gamma}{\omega_0 - \omega} + \frac{\Gamma}{\omega_0 + \omega}\right)^2 I(\mathbf{r}). \tag{1.66}$$

These equations show two resonance frequency for $\omega = \pm\omega_0$. Introducing the detuning $\Delta = \omega - \omega_0$, for the far-red regime it holds that $\Gamma \ll |\Delta|$. For instance, the D_2 line in Rb atoms has a transition frequency of $\sim$ 385 THz thus a trap typically works at few THz of detuning, while $\Gamma/(2\pi) \sim 6$ MHz. This allows the introduction of the well-known rotating-wave approximation, according to which the anti-resonant term can be neglected, further simplifying the equations to

$$U_{\text{dip}}(\mathbf{r}) = \frac{3\pi c^2}{2\omega_0^3}\frac{\Gamma}{\Delta} I(\mathbf{r}) \tag{1.67}$$

$$\Gamma_{\mathrm{sc}}(\mathbf{r}) = \frac{3\pi c^2}{2\hbar\omega_0^3}\left(\frac{\Gamma}{\Delta}\right)^2 I(\mathbf{r}) = \frac{\Gamma}{\hbar\Delta}U_{\mathrm{dip}} \tag{1.68}$$

Equations (1.67) and (1.68) are the fundamental equations governing the behavior of optical dipole traps and yield important physical considerations for their implementation. The sign of U_{dip} is determined by the sign of the detuning Δ. For red detuning ($\Delta < 0$) the potential is negative, confining atoms at intensity maxima, whereas for blue detuning ($\Delta > 0$), the potential is positive pushing atoms to intensity minima. Because the potential depth scales as I/δ while the scattering rate scales as I/Δ^2, optical traps are typically realized with high intensity field and large detunings, in this way, the trap depth is maximized while maintaining a low scattering rate.

1.2.2 Multi-level atoms

Moving from the two-level atom to a real atom, the electronic transitions become more complex, and the calculation of the dipole potential must take into consideration all the possible allowed transitions. This lead to a dipole potential that depends on the particular electronic substate.

1.2.2.1 Ground-state light shifts

Labelling as ε_i the energy levels of the multi-level atom, the effect of the laser light on the levels can be described by means of the non-degenerate time-independent second order perturbation theory. Considering the electric dipole interaction Hamiltonian $\mathcal{H}_I = -\hat{\mathbf{d}}\cdot\mathbf{E}$, the perturbative correction on the i-th energy level is

$$\Delta E_i = \sum_{j\neq i} \frac{|\langle j|\mathcal{H}_I|i\rangle|}{\varepsilon_i - \varepsilon_j}. \tag{1.69}$$

The calculation can be developed by employing the dressed atom approach, in which the quantum states describe the combined atom-field system. Defining the unperturbed atomic ground state energy as zero, in this framework an unperturbed dressed state has an energy $\varepsilon_i = n\hbar\omega$ that depends solely on the number of photons n in the field. The absorption of a photon transitions the system to another dressed state characterized by the energy $\varepsilon_j = \hbar\omega_j + \hbar(n-1)\hbar\omega$, where $\hbar\omega_j$ represent the bare atomic transition energy. In such a way, the denominator of equation (1.69) reduces to

$$\varepsilon_i - \varepsilon_j = -\hbar\omega_j + \hbar\omega = -\hbar\Delta_{ij}, \tag{1.70}$$

where Δ_{ij} is the detuning of the $i \rightarrow j$ transition with respect to the laser field.

Considering a two-level atom first, (1.69) has only one term, using equations (1.64) and

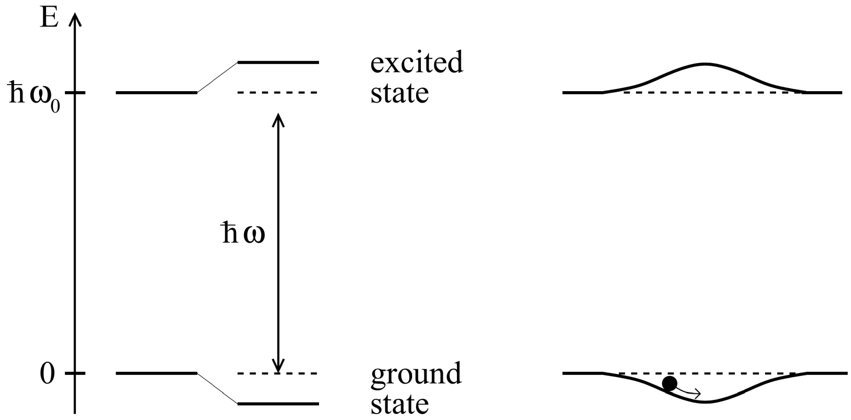


Figure 1.6: *From [5]. Left: light-shifted energy levels for a two-level atom for a red detuning ($\Delta < 0$). Right: a spatially inhomogeneous field produces a potential well that traps the atoms.*

(1.70) and $I = 2\epsilon_0 c|\mathcal{E}|^2$, one gets:

$$\Delta E = \pm\frac{|\langle e|\mu|g\rangle|^2}{\Delta}|E|^2 = \pm\frac{3\pi c^2}{2\omega_0^3}\frac{\Gamma}{\Delta}I, \tag{1.71}$$

for the ground and excited state (upper and lower sign, respectively). Equation (1.71) shows that the energy shift of the ground state (the solution with the sign +) exactly correspond to the dipole potential (1.67). This allows the reinterpretation of the semi-classical result obtain for the external degrees of freedom (position and momentum) in terms of the dynamic of the internal degrees of freedom. In presence of the laser field, the energy levels are shifted by a quantity proportional to the field intensity with a sign that depends on the sign of the detuning. If a low saturation regime is assumed ($I/\Delta \ll 1$), all the atom can be considered in the ground state, and the light-shifted ground state became the effective potential that govern the motion of the atoms which drives them toward region of minima of ΔE (Figure 1.6).

For a multi-level atom equation (1.69) contains many terms similar to equation (1.71), accounting for all the allowed transitions from any ground state sublevel $|g_i\rangle$ to any excited state $|e_j\rangle$. However, each of these terms depends on the dipole matrix element $d_{ij} = \langle g_i|\hat{d}|e_j\rangle$. Such quantity can be expressed as

$$d_{ij} = c_{ij}||d|| \tag{1.72}$$

where $||d||$ is the reduced matrix element given by (1.64) and c_{ij} are real transition coefficients that take into account the coupling strength of the two involved states $|g_i\rangle$ and $|e_j\rangle$, and depends on their angular momentum value [5][7].

Therefore, the energy shift (1.69) for the ground state $|g_i\rangle$ is

$$\Delta E_i = \frac{3\pi c^2 \Gamma}{2\omega_0^3} I \times \sum_j \frac{c_{ij}^2}{\Delta_{ij}} \tag{1.73}$$

and according to the interpretation given from the two-level atoms, the dipole potential at low saturation is

$$U_{\text{dip}} = \Delta E_i. \tag{1.74}$$

Thus, the dipole potential in a multi-level atom arises from the contribution of all the allowed transition, each weighted by the ratio between its line strengths c_{ij}^2 and its detuning Δ_{ij}.

1.2.2.2 Fine and Hyperfine structures contribution for Alkali atoms

Many laser cooling experiments utilize alkali metal atoms. This preference arises because their principal transition frequencies fall within the visible and infrared (VIS-IR) range making them easily accessible with standard laser technologies. Moreover, their relatively simple atomic structure are particularly convenient, allowing easy identification of closed transitions necessary for continuous cooling cycles, while minimizing the number of repumper lasers needed to address the population leaking to dark states.

When considering the fine structure arising from the spin-orbit interaction, the electronic ground state of alkali atoms is ${}^2S_{1/2}$. From this state, the electric dipole selection rules ($\Delta L = \pm 1$, $\Delta S = 0$) allow for two primary transitions, known as $D-$lines:

- D_1: ${}^2S_{1/2} \to {}^2P_{1/2}$;
- D_2: ${}^2S_{1/2} \to {}^2P_{3/2}$.

Beyond the fine structure, the energy levels undergo further splitting due to the coupling between the nuclear spin and the total electronic angular momentum. The resulting configuration is called hyperfine structure. Denoting the fine structure splitting as $\hbar\Delta_{FS}$, the ground state hyperfine splitting as $\hbar\Delta_{HFS}$, and the excited state hyperfine splitting as $\hbar\Delta'_{HFS}$, these characteristic energy scales obey $\Delta_{FS} \gg \Delta_{HFS} \gg \Delta'_{HFS}$. The complete electronic energy level scheme for an alkali atom with nuclear spin $I = 3/2$ is shown in figure 1.7.

Assuming that all optical detunings stay large compared with the excited-state hyperfine splitting Δ'_{HFS}, equation (1.73) can be calculated for an alkali atom in the ground state of total angular momentum F and magnetic quantum number m_F, obtaining [5]:

$$U_{\text{dip}}(\mathbf{r}) = \frac{\pi c^2 \Gamma}{2\omega_0^3} \left(\frac{2 + \mathcal{P} g_F m_F}{\Delta_{2,F}} + \frac{1 - \mathcal{P} g_F m_F}{\Delta_{1,F}} \right) I(\mathbf{r}), \tag{1.75}$$

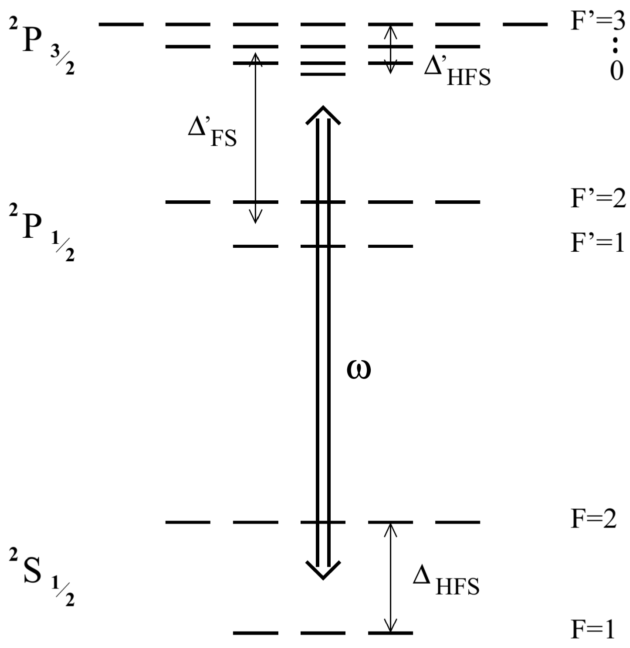


Figure 1.7: *From [5]. Energy level scheme of an alkali atom with $I = 3/2$ such as ^{87}Rb.*

where the two term in the parentheses represent respectively the contribution of the D_2 and D_1 line. Here, g_F is the Landé factor, $\mathcal{P}$ characterizes the laser polarization ($\mathcal{P} = 0, \pm 1$ for linear polarization and circular $\sigma^\pm$ respectively), and $\Delta_{2,F}$ and $\Delta_{1,F}$ denote the detunings of the laser field relative to the transitions from the $^2S_{1/2}, F$ ground state to the center of respectively the $^2P_{3/2}$ and $^2P_{1/2}$ hyperfine excited state.

If the detunings are large with respect to the fine-structure splitting ($|\Delta_{1,F}|, |\Delta_{2_F}| \gg \Delta'_{FS}$), introducing the detuning Δ with respect to the center of the $D-$line doublet, expanding equation (1.75) at first order of the small parameter Δ'_{FS}/Δ, one gets [5]:

$$U_{\text{dip}}(\mathbf{r}) = \frac{3\pi c^2}{2\omega_0^3} \frac{\Gamma}{\Delta} \left(1 + \frac{1}{3} \mathcal{P} g_F m_F \frac{\Delta'_{\text{FS}}}{\Delta}\right) I(\mathbf{r}). \tag{1.76}$$

The zeroth order term corresponds exactly to the dipole potential of a simple two-level atom (see Eq. (1.67)). The first-order term introduces a small, spin-dependent correction depending on the laser polarization $\mathcal{P}$ and the magnetic quantum number m_F.

Physically, equation (1.76) describes that, when the fine structure is completely unresolved due to the large laser detuning, the atom effectively acts as a generic two-level system undergoing an $s \to p$ transition. Consequently, the resulting dipole potential is dominated by the scalar term, matching the two-level model previously discussed.

Since in the unresolved fine structure case, the polarization-dependent term is heavily suppressed by the small parameter Δ'_{FS}/Δ (and vanishes entirely for linearly polarized light), the atom effectively acquires only the scalar light shift of the unperturbed s state. Consequently, all hyperfine and magnetic sublevels experience a uniform trapping potential that confines the atom while preserving its internal spin degeneracy.

In a more general case, when the fine structure is resolved, but the hyperfine structure is unresolved ($|\Delta_{1,F}|, |\Delta_{2,F}| \gg \Delta_{HFS} \gg \Delta'_{HFS}$), only linearly polarized light guarantees that the dipole potential becomes completely independent of m_F and F [5]. Therefore, linearly polarized light is usually preferred for dipole trap implementation.

1.2.3 Heating mechanism

Typical trap depths for optical dipole traps are usually below 1 mK. Therefore, capturing a substantial number of atoms requires a preliminary cooling stage. For instance, the experiment discussed in this thesis employs a magneto-optical trap (MOT) followed by an optical molasses phase.

Given these shallow confining potentials, even minor heating rates can largely limit the trap lifetime. It is therefore important to outline the mechanisms responsible for the parasitic heating induced by the trapping light.

1.2.3.1 Heating source

When atoms interact with laser light, fluctuations in the spontaneous emission and absorption always causes heating (this is also the phenomenon that puts a lower limit to cooling mechanism [8][7]). For the optical dipole trap the atoms absorbs photons and re-emits them in random directions (isotropic radiation), statistical oscillations in both these processes cause heating. The absorption of photons from the dipole light is associated with a recoil energy $E_{\text{rec}} = k_B T_{\text{rec}}/2$ per scattering event and happens only in the direction of the laser beam, hence it is anisotropic. On the other hand, the re-emission scattering events are also associated with a recoil energy E_{rec}, however the emission directions are random, thus the effect is spread over all the three spatial direction. Thus, for a single cycle the atom receives E_{rec} from the absorption and then E_{rec} from the spontaneous emission, for a total of $2E_{\text{rec}}$, in a time $\bar{\Gamma}_{\text{sc}}^{-1}$.
It leads to the total heating power

$$P_{\text{heat}} = 2E_{\text{rec}}\bar{\Gamma}_{\text{sc}} = k_B T_{\text{rec}}\bar{\Gamma}_{\text{sc}}, \tag{1.77}$$

the heating depends on the mean scattering rate $\bar{\Gamma}_{\text{sc}}$ and on the recoil E_{rec} which depends on the laser frequency.

Other fundamental source of heating, such as the redistribution of photons between different travelling wave, should be quenched by the far-detuned working frequency [5].

1.2.3.2 Heating rate

Assuming that the trapped atoms are at thermal equilibrium, the mean energy can be expressed through the energy equipartition theorem. Assuming the trap as a three-dimensional harmonic potential well, there is a contribution of three quadratic degrees of freedom from the kinetic energy, and other three from the harmonic potential. According to the energy partition theorem, the mean energy is

$$\bar{E} = 3k_B T. \tag{1.78}$$

Using (1.77) in (1.78), one can obtain an equation that describes the temperature change with time:

$$\dot{T} = \frac{1}{3} T_{\text{rec}} \bar{\Gamma}_{\text{sc}}. \tag{1.79}$$

Expressing the dipole potential as

$$\bar{U}_{\text{dip}} = U_0 + \bar{E}_{\text{pot}} = U_0 + \frac{3}{2} k_B T, \tag{1.80}$$

where U_0 is the potential energy minimum.
Using equation (1.68), $\bar{U}_{\text{dip}}$ can be expressed as:

$$\bar{U}_{\text{dip}} = \frac{\hbar \Delta \Gamma_{\text{sc}}}{\Gamma}. \tag{1.81}$$

Substituting this into Equation (1.80) yields the mean scattering rate:

$$\bar{\Gamma}_{\text{sc}} = \frac{\Gamma}{\hbar \Delta} (U_0 + \frac{3}{2} k_B T). \tag{1.82}$$

It is important to notice that, for a blue-detuned trap, $U_0 = 0$ and the trap depth $\hat{U}$ is given by the maximum value of the potential surrounding the minimum, by contrast for a red-detuned trap $\hat{U} = |U_0|$. Finally, using equation (1.82) in (1.79), with a depth $\hat{U} \gg k_B T$, one gets the following temperature changing rate for a red and a blue trap:

$$\dot{T}_{\text{red}} = \frac{1}{3} T_{\text{rec}} \frac{\Gamma}{\hbar |\Delta|} \hat{U}, \tag{1.83a}$$

$$\dot{T}_{\text{blue}} = \frac{1}{2} T_{\text{rec}} \frac{\Gamma}{\hbar \Delta} k_B T. \tag{1.83b}$$

Equations (1.83) describe the increasing in temperature of the atomic cloud if subjected to the dipole trap only. It highlights another blue trap limitation: while a red trap has a linear heating rate (as long as $\hat{U} \gg k_B T$), the blue trap shows an exponential heating law.

Chapter 2

Materials And Method

This chapter describes the experimental apparatus used in this work. First, a brief overview of the implementation of the Magneto-Optical Trap (MOT) for rubidium-87 atoms (^{87}Rb) is presented (this setup was already available prior to the present thesis work). The chapter then focuses on the work carried out during this thesis, namely the implementation of the Optical Dipole Trap setup employed to transfer and trap atoms from the MOT.
Although beyond the scope of the current work, it is worth noting that the long-term objective of this experiment is to confine the atoms within the core of a hollow-core fiber.

2.1 Magneto-optical trap

2.1.1 The Vacuum Chamber

2.1.2 The laser system

2.1.2.1 Master laser

2.1.2.2 Cooler and repumper lasers

2.2 Optical dipole trap

This section describes the implementation of the Optical Dipole Trap (ODT) setup used to transfer atoms from the MOT into a conservative optical potential. The ODT beam is guided through a Hollow-Core Photonic Crystal Fiber (HCPCF). As mentioned in Section 2.1, the HCPCF integrated into the experimental apparatus was already present prior to this thesis work. However, a test HCPCF was employed during the development of the setup in order to optimize the alignment procedure.
During the implementation of the system, dedicated diagnostic tools were also developed to investigate the near-field beam profile and improve the fiber-coupling alignment. The following sections describe the ODT optical configuration and the experimental procedures adopted for the alignment and loading of the trap.

2.2.1 1064nm laser characterization

The ODT is realized by using an IPG Photonics YLR-20-1064-LP-SF laser, a linearly polarized Ytterbium fiber laser that emits 1064 nm light, at a maximum nominal power of 20 W. For this laser model, the beam shape change significantly upon changes of the nominal power set. Therefore, the mode profile has been characterized as a function of the power set in order to allow a correct realization of the HCPCF injection setup. This calibration also provides a useful reference for future optimization of the HCPCF injection.

2.2.1.1 Methodology

To characterize the beam properties, its spatial profile is modeled as a Gaussian beam. Therefore, the beam intensity distribution can be described by [3]:

$$I(r,z) = I_0 \left(\frac{w_0}{w(z)}\right)^2 \exp\left(\frac{-2r^2}{w(z)^2}\right), \tag{2.1}$$

where

- z is the axial distance from the beam waist;
- r is the radial distance from the center of the beam;
- $w(z)$ is the beam radius, defined as the radius at which the intensity decreases of a factor $1/e^2$ with respect to the intensity at the beam center;
- $w_0 = w(0)$ is the beam waist, i.e. the beam radius at its focus;
- I_0 is the intensity at the center of the beam at its waist.

The transverse intensity profile follows a Gaussian shape, hence the name of Gaussian beam.

At any axial position z, the beam radius is given by [3]:

$$w(z) = w_0 \sqrt{1 + \left(M^2 \, \frac{z}{z_R}\right)^2} \tag{2.2}$$

where

$$z_R = \frac{\pi w_0^2}{\lambda} \tag{2.3}$$

is the Rayleigh range, which represent the characteristic distance around the waist over which the beam remains approximately collimated, and $M^2 \geq 1$ is the beam quality factor, accounting for deviations from an ideal Gaussian beam. By acquiring intensity images at multiple longitudinal positions and extracting the corresponding beam radii, Equation (2.2) can be fitted to determine both the waist radius and the waist position.

The experimental setup used for the intensity measurements corresponds to the initial section of the ODT injection setup (see Section 2.2.2.1). With reference to Figure 2.3, measurements were performed on the beam exiting the reflected branch of the second polarization beam splitter (PBS2). The power was adjusted through the regulation of the half-wave plates HWP2 and HWP3, together with the addition of optical density filters when needed. A Complementary Metal-Oxide-Semiconductor (CMOS) camera was employed to take 17 intensity images separated by 25 mm along the propagation axis for the nominal laser powers of 4 W, 7 W, 10 W, 13 W, 16 W and 19 W.

The straightforward approach of extracting the beam radius from each image consists of fitting a 2D asymmetric Gaussian profile and extract the radius along the x and y axis as

$$w_x(z) = 2\sigma_x, \qquad w_y(z) = 2\sigma_y. \tag{2.4}$$

However, fitting equation (2.2) using these extracted radii led to values of $M^2 < 1$ for all the power settings except 4 W, which is unphysical. However, enforcing the constraint $M^2 \geq 1$ resulted in visibly poor agreement between the fitted curves and the measured data. A likely explanation is that the presence of several optical components before the camera introduce aberrations or beam clipping that make equation (2.2) incorrect.

For this reason a second method called $D4\sigma$ was tested [9]. In this method, the beam radius is determined from the second moment of the beam intensity distribution with respect to the intensity centroid. For the x-axis:

$$\hat{\sigma}_x^2 = \frac{\int_{-\infty}^{\infty} (x - x_0)^2 I(x, y) \, dx \, dy}{\int_{-\infty}^{\infty} I(x, y) \, dx \, dy} \tag{2.5}$$

where x_0 is the intensity centroid x-coordinate, and $I(x, y)$ is the beam intensity profile.

Equation (2.2) is then fitted using the radii

$$w_x = 2\hat{\sigma}_x, \qquad w_y = 2\hat{\sigma}_y \,. \tag{2.6}$$

The $D4\sigma$ method should be more robust than the Gaussian fitting when dealing with non-ideal beam profiles, since it accounts for intensity contributions in the beam tails that may not be properly captured by a Gaussian fit. However, the $D4\sigma$ is very sensitive to anisotropic background noise, which can bias both the centroid and second-moment calculations. In the present setup, the presence of the PBS1 and the beam dump that dissipates a large quantity of optical power, generates a non-uniform scattering background. This asymmetry introduces systematic distortions in the $D4\sigma$ analysis. Consequently, even this method did not consistently yield $M^2 \geq 1$ without constraints, although it performed better than the Gaussian fit, giving naturally $M^2 \geq 1$ for 4 W, 16 W and 19 W.

For both analysis methods, a preliminary Gaussian fit was first performed to find approximately the beam center position. A region of interest (ROI) extending to 5σ around the beam center was then selected. Subsequently, either a refined Gaussian fit was performed or the centroid and second moments were calculated to determine the beam radius before fitting Equation (2.2).

2.2.1.2 Results

The beam waist radii and waist positions obtained with the two analysis methods are summarized in Tables 2.1 and 2.2. The fit of Equation (2.2) was performed using the Trust Region Reflective (TRF) algorithm while imposing the constraint $M^2 \geq 1$. The final reported values were obtained by averaging the results of the Gaussian fit and $D4\sigma$ methods.

The waist radius as a function of the nominal laser power is plotted in Figure 2.1 top. Except for 4 W and 7 W, the results reveal a small astigmatism, with the waist measured along the x direction exceeding that along the y direction by approximately 1–2%. Furthermore, the graph shows that the waist radius decreases approximately linearly from 7 W to 19 W, indicating a significant power dependence of the laser mode size.

The waist longitudinal position as a function of the nominal laser power is plotted in Figure 2.1 bottom. These values should be interpreted with caution. Imposing $M^2 \geq 1$ allow the fits to better match the middle and far points which depends asymptotically on solely w_0 (limit $z \gg 1$ in equation 2.2), but strongly miss the prediction on the point at short distance that give the information on the waist position (see Figure 2.2). Although the extracted values carry significant systematic uncertainty, the observed trend appears physically plausible, with the beam waist shifting downstream as the nominal laser power increases from 4 W to 13 W.

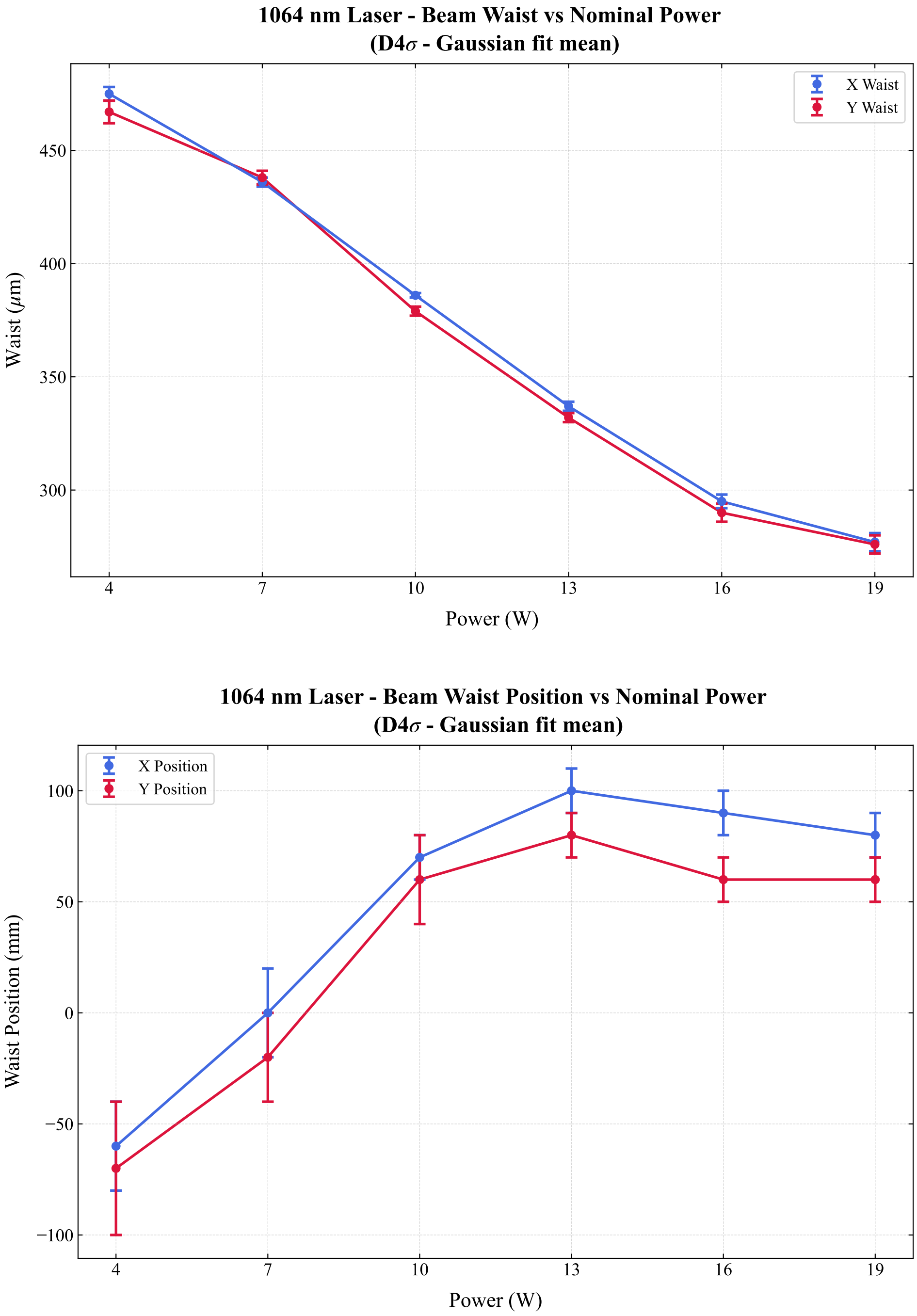


Figure 2.1: *Characterization of the 1064 nm laser beam parameters as a function of nominal output power. (Top) Transverse beam waist size versus power. (Bottom) Longitudinal position of the waist. The reported values correspond to the average of the results obtained using the D4σ and Gaussian-fit methods. Blue and red markers denote measurements along the orthogonal x and y transverse axes, respectively.*

Table 2.1: *Comparison of the beam waist w_0 obtained via the D4σ method and the Gaussian fit method. The final values correspond to the average of the two estimates.*

Nominal Power (W)	Estimated waist w_0 (μm)					
	D4σ-x	*D4σ-y*	Gaus-x	Gaus-y	Avg-x	Avg-y
4	481 ± 5	474 ± 7	469 ± 4	460 ± 7	475 ± 3	467 ± 5
7	454 ± 3	452 ± 4	419 ± 3	423 ± 5	436 ± 2	438 ± 3
10	403 ± 2	394 ± 3	370 ± 2	364 ± 3	386 ± 1	379 ± 2
13	351 ± 2	343 ± 3	323 ± 4	321 ± 4	337 ± 2	332 ± 2
16	307 ± 2	298 ± 4	283 ± 6	282 ± 6	295 ± 3	290 ± 4
19	288 ± 3	284 ± 4	266 ± 7	268 ± 6	277 ± 4	276 ± 4

Table 2.2: *Comparison of the beam waist position z_0 extracted via the D4σ method and the Gaussian fit method. The final values correspond to the average of the two estimates.*

Nominal Power (W)	Estimated waist position z_0 (mm)					
	D4σ-x	*D4σ-y*	Gaus-x	Gaus-y	Avg-x	Avg-y
4	-90 ± 40	-80 ± 50	-40 ± 20	-60 ± 40	-60 ± 20	-70 ± 30
7	20 ± 30	-3 ± 40	-20 ± 20	-40 ± 20	0 ± 20	-20 ± 20
10	90 ± 20	70 ± 30	50 ± 10	40 ± 10	70 ± 10	60 ± 20
13	110 ± 10	90 ± 20	90 ± 20	70 ± 20	100 ± 10	80 ± 10
16	72 ± 9	50 ± 20	100 ± 20	80 ± 20	90 ± 10	60 ± 10
19	70 ± 10	40 ± 10	90 ± 20	70 ± 20	80 ± 10	60 ± 10

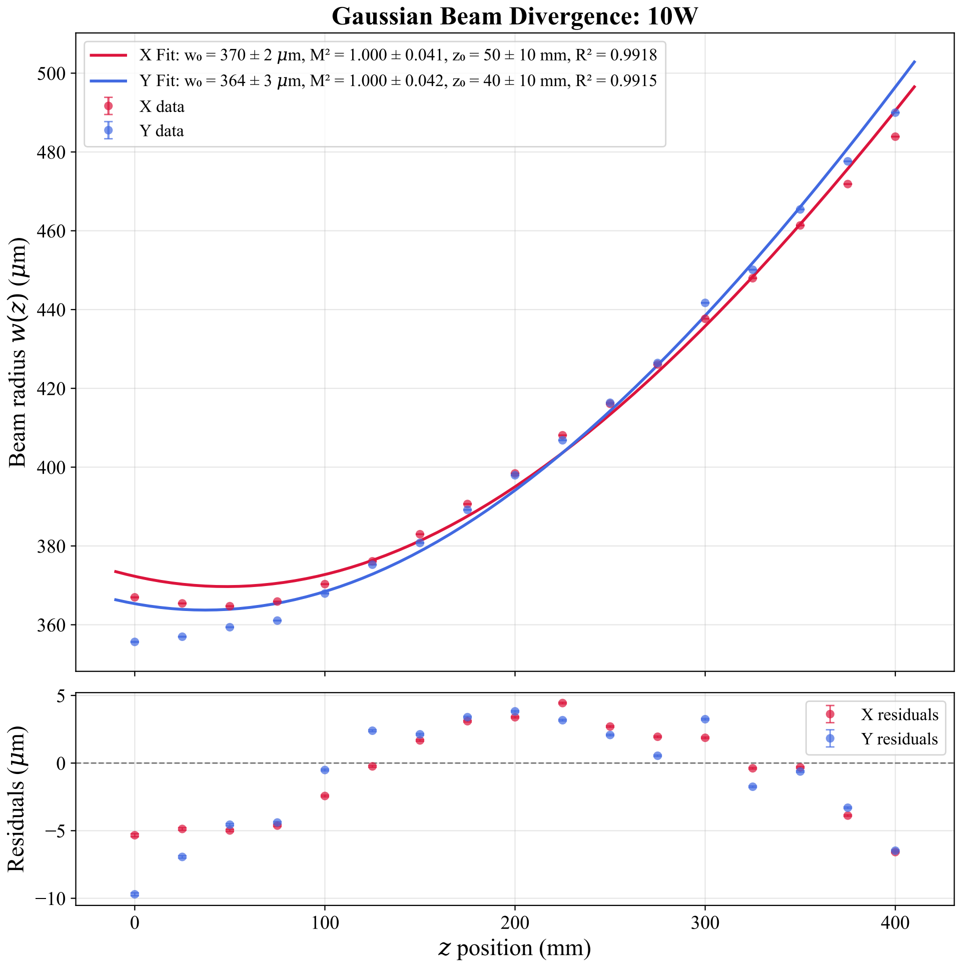


Figure 2.2: *Gaussian beam divergence profile measured at a nominal laser power of 10 W. The beam radius $w(z)$ is shown as a function of the longitudinal position z along the x (red) and y (blue) axes. Solid lines denote Gaussian propagation fits performed under the constraint $M^2 \geq 1$. The systematic discrepancy observed at short propagation distances ($z < 100mm$) results from the imposed M^2 constraint, which prioritizes agreement with the intermediate- and far-field data at the expense of the near-field region. Consequently, the accuracy of the fitted beam waist position z_0 is reduced.*

2.2.2 Optical setup

In this experiment the ODT light is guided through a HCPCF, which introduces a critical complication in the injection procedure. Indeed, in contrast with standard telecom fibers, HCPCFs are multi-mode fibers. As a consequence, even small deviations of the injected beam from the ideal injection angle excites higher order modes in addition to the fundamental one. This results in distorted output beam profiles, which are not well suited for the realization of an optical conveyor belt based on the superposition of the beam guided by the HCPCF and a beam guided through a conventional telecom fiber supporting only the fundamental LP_{01} mode.
For this reason, particular cure was devoted to the alignment procedure. A significant improvement in the quality of the transmitted mode was achieved by optimizing the injection while inspecting the near-field of the transmitted beam, rather than simply maximizing the transmission efficiency. The following sections describe the injection setup and the horizontal microscope built for the near-field inspection.

2.2.2.1 Injection setup

Figure 2.3 shows the optical setup scheme used for the injection of the laser light in the HCPCF. The scheme is composed of the following elements:

- **Half Wave Plate 1** (HWP1) and **Optical Isolator**. The optical isolator prevents back reflections from propagating towards the laser cavity. High-power lasers are particularly sensitive to back reflections, which can introduce power instabilities and potentially damage the laser system. Since optical isolators operate correctly only for a specific input polarization, HWP1 is used to optimize the polarization of the incoming beam and maximize the transmission through the isolator.

- **HWP2** and **Polarization Beam Splitter 1** (PBS1). These components are used to regulate the optical power delivered to the subsequent stages of the setup. The unwanted fraction of the beam is directed onto a beam dump. This configuration is particularly useful during alignment procedures, where the optical power is reduced to a few milliwatts for safety reasons.

- **HWP3** and **PBS2**. PBS2 splits the beam into two different branches. The first is employed for the HCPCF injection, while the second will be injected into a PANDA polarization-maintaining fiber, which eventually will provide the second beam for the realization of the optical conveyor belt. HWP3 is adjusted to have approximately 50/50 power splitting ratio, although this value can be modified if required.

- **Acousto-Optic Modulator** (AOM). The AOM is employed as a fast switch for the ODT beam. The device operates using an acoustic wave at 110 MHz, generated by a

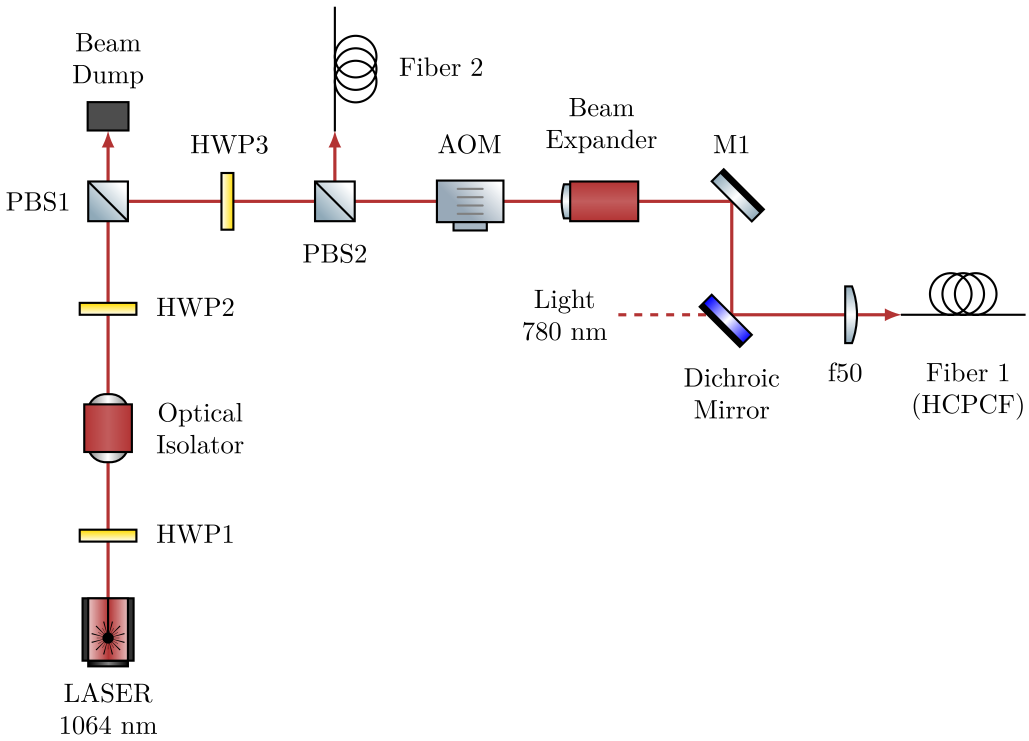


Figure 2.3: *Scheme of the optical setup for 1064 nm ODT beam injection into the HCPCF. The Acousto-Optic Modulator (AOM) operates at 110 MHz, and the subsequent setup utilizes the fist diffraction order. The zero-th order is blocked by a beam dump (not drawn). The 780 nm light setup is not drawn, its presence will be important in the future stages of the experiment for the characterization of the atoms inside the HCPCF core. Fiber 2 injection setup is not drawn as well.*

Direct Digital Synthesizer (DDS), causing the first diffraction order to be frequency shifted by the same amount. Although this shift is negligible with respect to the trap depth of the ODT, it is essential for the realization of the optical conveyor belt.

- **Beam expander**. A ∼3X beam expander is employed to increase the beam diameter and collimate it before the focusing conditions for coupling into the fiber.

- **Mirror 1** (M1) and **Dichroic Mirror**. Both mirrors are mounted on tilt stages and their angular adjustment is used for optimizing the fiber injection alignment. The dichroic mirror also combines the 1064 nm ODT beam with a 780 nm beam that will be used for the characterization of the atoms trapped inside the HCPCF core.

- **Focusing Lens** (f50). A lens with focal length $f = 50$ mm focuses the beam into the HCPCF. It is mounted on a horizontal translation stage, providing an additional degree of freedom for optimizing the injection alignment.

- **Fiber 1** (HCPCF). The HCPCF is mounted on an x-y translation stage and terminated with an FC/APC connector. The translation stage allows transverse displacement of the fiber, providing additional alignment degrees of freedom during the injection procedure, although these degrees of freedom were found to have limited impact on the final optimization.

Using the beam waist measurements in Section 2.2.1.2, a nominal laser power of 10 W was selected as the optimal operating point. Since the fitted waist positions were found to be affected by large systematic issues, the beam waist was assumed to be located at the output face of PBS2. The radius at the entrance of the beam expander can be estimated using (2.2), and the measured distance of approximately 175 mm between PBS2 and the beam expander (see figure 2.5). Assuming an ideal beam $M^2 = 1$, using the x waist value for 10 W of $w_0 = 384\ \mu$m, the beam radius at the entrance of the beam expander is estimated to be $w(\text{beam expander entrance}) \simeq 414\ \mu$m.
The beam expander provides a magnification factor of three, yielding $w(\text{beam expander exit}) \simeq 1240\ \mu$m. At this stage the corresponding Rayleigh range, calculated using 2.3 is $z_R \simeq 4.5$ m. Since the remaining propagation distance before the injection optics is negligible compared to the Rayleigh range, the beam is considered approximately collimated. Under this assumption, the beam waist after the injection lens can be estimated using [3]:

$$w_{\text{out}} = \frac{\lambda f}{\pi w_{\text{in}}}, \tag{2.7}$$

where $f = 50$ mm is the focal length of the injection lens. Propagating the uncertainty, one gets:

$$w_{\text{inj}} \simeq (13.6 \pm 0.1)\ \mu\text{m}.$$

This result is obtained in the ideal case, the real waist at the injection is expected to be closer to the target value of approximately $\sim 18\ \mu$m. For instance, assuming a more realistic beam quality factor of $M^2 = 1.2$ leads to

$$\tilde{w}_{\text{inj}} \simeq (15.1 \pm 0.1)\ \mu\text{m}$$

Additional deviations from the ideal prediction are expected due to the AOM, which distorts the beam profile by compressing it along the y direction and expanding it along the x direction, resulting in an expected beam radii ratio $w_y/w_x \approx 0.85$ at AOM exit, which is reversed at injection by the relation (2.7).
To verify these predictions, the beam profile after the injection lens was characterized following a procedure analogous to that described in Section 2.2.1, yielding

$$w_{\text{inj-}x} = (17.1 \pm 0.7\ \mu\text{m}) \qquad w_{\text{inj-}y} = (19.3 \pm 0.4\ \mu\text{m}),$$

sufficiently close to the target waist.

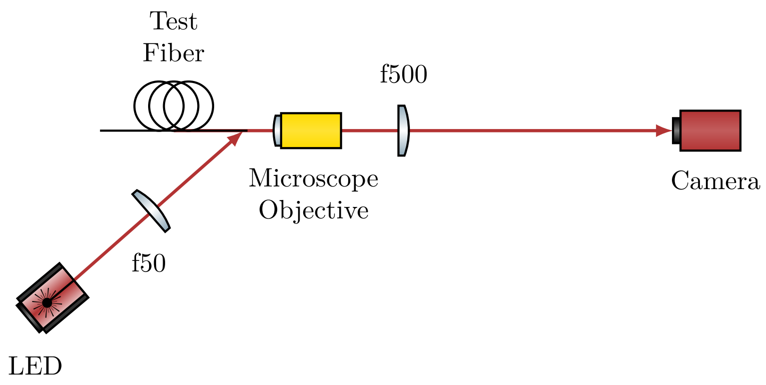


Figure 2.4: *Scheme of the optical setup for the horizontal microscope.*

2.2.2.2 Horizontal microscope

Initially, the injection into the HCPCF installed inside the vacuum chamber was attempted by using a Complementary Metal-Oxide-Semiconductor (CMOS) camera to monitor the beam profile. However, despite the effort, strongly deformed mode profiles were obtained.

In order to better investigate the injection procedure and simplify the alignment process, a separate test HCPCF was employed outside the vacuum chamber. The test fiber consisted of an approximately 50 cm HCPCF segment with the same FC/APC connector as the fiber in the chamber. The test fiber allowed an easy measurement of the efficiency on the exit facet of the beam without any window in the path, and being outside the vacuum chamber allow for easier mode imaging. Moreover, since the two fibers shared the same connector, the optimal alignment conditions identified with the test fiber were expected to provide a good starting point for the alignment of the HCPCF installed inside the chamber. Nevertheless, even with the test fiber, optimization based solely on transmission efficiency and far-field inspection still resulted in poor beam quality.

The fiber manufacturer suggested that the optimal injection condition should be determined by inspecting the shape of the near-field of the transmitted mode. For this purpose, a horizontal microscope system was developed to inspect the HCPCF facet and to investigate the near-field profile of the transmitted mode. The optical scheme of the microscope is shown in Figure 2.4.

The fiber was positioned inside a 90° V-groove of depth 350 μm. The fiber was fixed inside the groove by means of a thin foam layer attached to a tiltable mount using adhesive tape. The coating was removed from the fiber end, leaving approximately 15 mm of suspended fiber in order to allow lateral illumination of the facet.

The HCPCF facet was illuminated by a red LED (wavelength $\lambda = 635$ nm) focused by an $f = 50$ mm lens onto the final suspended portion of the fiber. Part of the red light propagated through the silica structure forming the kagome lattice, making them visible on the microscope camera. The observation of the kagome structure was used to optimize

the microscope focus and to inspect the integrity and cleanliness of the fiber facet.
The light emerging from the fiber was collected by an optical microscope objective lens (Olympus PLN10X objective, nominal focal length $f = 18$ mm, working distance $d = 10.6$ mm) and subsequently re-imaged onto a CMOS camera by means of an $f = 500$ mm lens. The expected magnification is $M = f_{\text{tube}}/f_{\text{lens}} \approx 27.8$. The objective is mounted on a horizontal translation stage used for the focussing. The camera was connected to a computer for real-time imaging of the transmitted near-field mode. In addition, a powermeter could be inserted between the fiber output and the objective lens in order to evaluate the coupling efficiency by comparing the transmitted power with the optical power measured before the injection stage. Finally, a black paper tube was placed between the imaging $f = 500$ mm lens and the camera to reduce background light from the LED illumination and from ambient room light.
The alignment procedure was carried out by optimizing the transmitted near-field profile in order to obtain a mode as close as possible to a single-lobed structure, results and further details are found in Section 2.2.3.
Figure 2.5 shows the top-down photographs of the complete experimental apparatus, including both the injection setup and the horizontal microscope system, and Figure 2.6 shows a detailed view of the horizontal microscope.

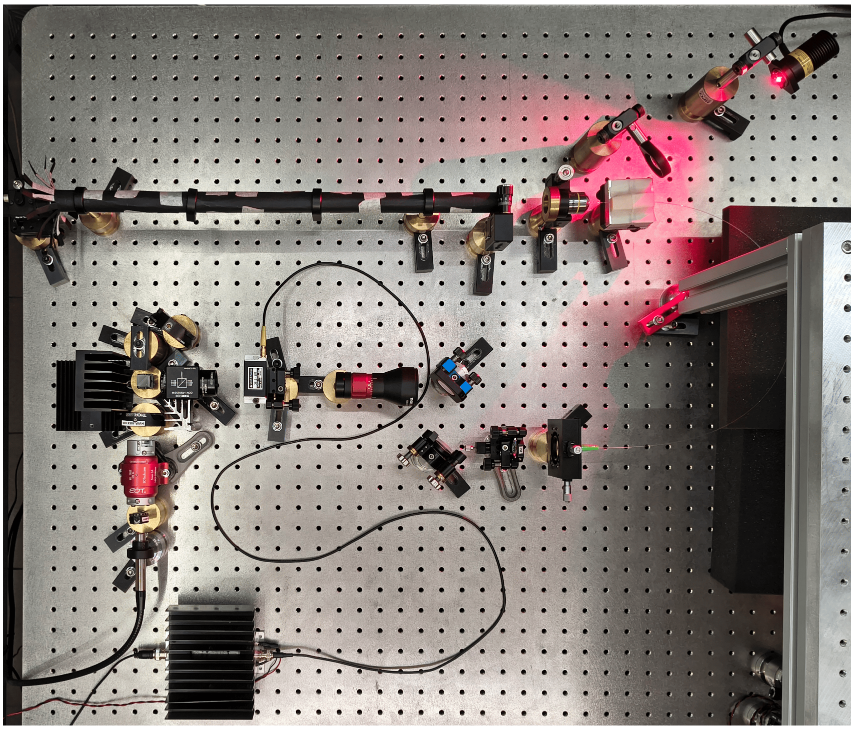


Figure 2.5: *Top-down photograph of the complete experimental apparatus. The horizontal microscope, shielded by the black paper tube, is visible at the top of the image. The optical injection setup is located in the center, while the RF amplifier driving the AOM is placed at the bottom. For reference, the distance between the screw holes on the breadboard is 25 mm.*

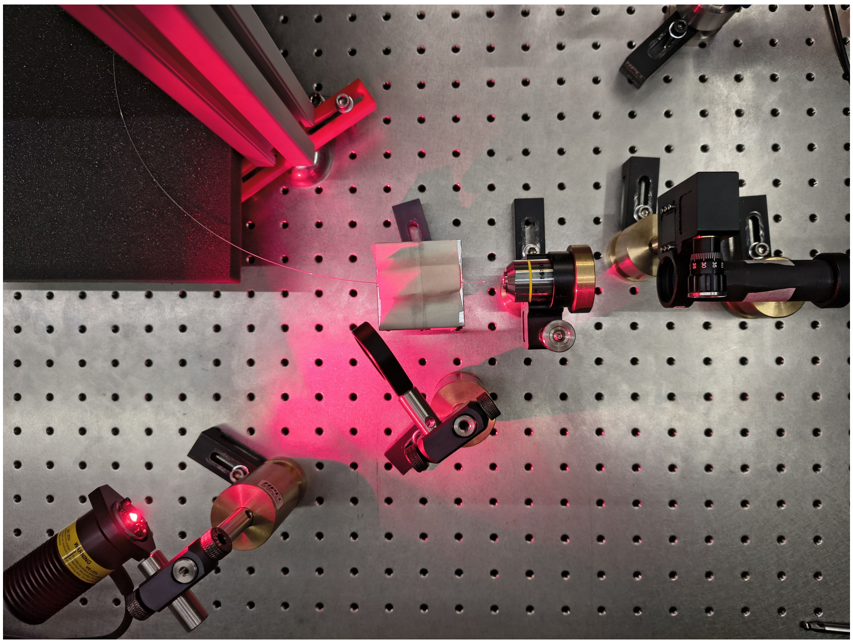


Figure 2.6: *Detailed top-down view of the horizontal microscope. The setup highlights the test fiber, the red LED providing lateral illumination to the fiber facet, the microscope objective lens and the refocusing lens.*

2.2.2.3 Chamber fiber optical imaging setup

After the optimization and the validation of the injection alignment procedure using the test fiber, the same approach was applied to the fiber inside the vacuum chamber.

To investigate the near-field profile, a dedicated optical imaging setup was build on a breadboard positioned above the vacuum chamber (Figure 2.7). Since the optical fiber is located inside the chamber, a first stage microscope was designed to create a virtual image of the fiber facet outside the chamber. For this purpose a f150 lens (f = 150 mm) was placed approximately 150 mm above the fiber facet, directly above the chamber window, such that the light emerging from the fiber is collimated. The collimated beam propagate through an opening and is redirected horizontally by a mirror (M1) tilted at 45°. A second lens f300 (f = 300 mm) refocuses the beam, forming an inverted image of the fiber facet with a magnification of $\times 2$.

A second imaging stage, consisting of lenses L3 and L4, acted as a microscope for this intermediate image and projected a magnified image onto the sensor of a camera. The focal lengths of L3 and L4 were modified several times throughout the experiment to optimize the field of view and magnification. The configuration used most frequently employed an L3 lens with f = 100 mm and an L4 lens with f = 500 mm, resulting in an additional magnification of $\times 5$. Combined with the first imaging stage, this yielded an overall magnification of approximately $\times 10$, which was used for the majority of the measurements.

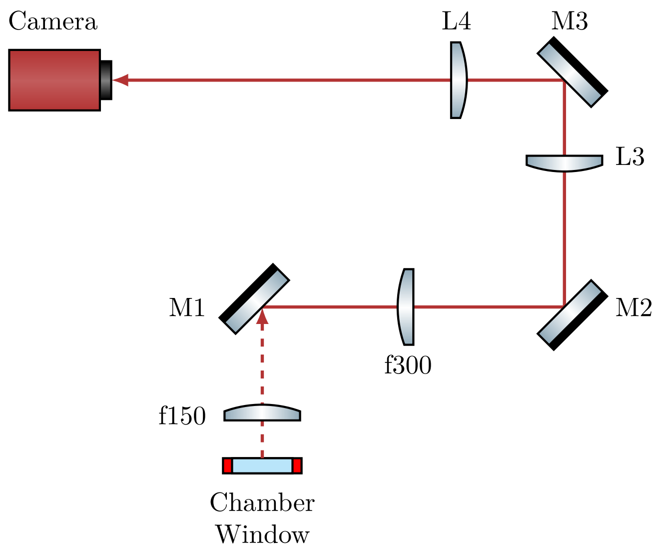


Figure 2.7: *Scheme of the optical setup used to image the fiber inside the vacuum chamber. The optical path between the chamber window and the mirror M1 is vertical (characterized by the dashed beam line), whereas the rest of the setup is arranged horizontally on a breadboard mounted above the chamber. The focal length of lenses L3 and L4 were varied several times during the experiment. Most images where acquired using an* $f = 100$ mm *and an* $f = 500$ mm *providing an overall magnification of approximately* $\times 10$ *combined with the first imaging stage.*

2.2.3 Injection alignment and near-field imaging

For a conventional single-mode telecom fiber, the alignment procedure is straightforward: it is sufficient to inject a low-power beam of few milliwatts, monitor the transmitted power at the fiber exit, and adjust the injected beam to maximize the guided optical power.

By contrast, aligning the injection of a multimodal HCPCF is significantly more complex. The silica jacket that surrounds the kagome core can guide the light with coupling efficiencies as high as 60%, although with badly shaped mode (see figure 2.8). Considering also that the jacket is large compared to the core, monitoring just the coupling efficiency is particularly ineffective, as the optimization can easily get trapped in a local maximum associated with jacket guiding.
As suggested by the manufacturer, the best procedure for the injection alignment resulted to be monitoring the near-field mode shape at the fiber exit. The complete alignment procedure begins with the use of a powermeter to ensure the impinging beam hits the fiber, even on the jacket. Then switch to live monitoring of the near-field with a camera and optimize its shape by tuning the degrees of freedom of the injection setup. Coupling efficiency is measured only after the mode profile has been optimized.

Applying this protocol, the test fiber injection was optimized employing the injection optical setup and the horizontal microscope described in Section 2.2.2. This approach yielded to a good result and thus was subsequently applied to align the injection into the

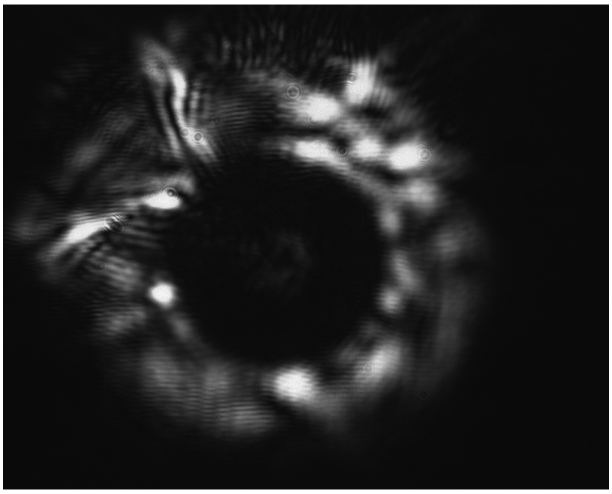


Figure 2.8: *Example of the near-field of a mode guided by the test fiber jacket. The light is mainly guided by the silica jacket surrounding the kagome structure, while only a weak signal is visible in the core. Note that this strongly deformed mode had a coupling efficiency of* $\sim 60\%$.

fiber in the vacuum chamber.

2.2.3.1 Test fiber

The test fiber was cleaved prior to inspection. Since the horizontal microscope has a sufficient resolution to resolve the kagome structure of the fiber, it was exploited for the focusing procedure. By monitoring the camera imaging in real-time, the kagome structure was focussed using the translation stage of the microscope objective. The objective refractive index can be considered roughly constant across the visible spectrum and into the near-infrared (NIR). This guarantees that when the structure (illuminated by red light) is in focus, the guided mode (1064 nm NIR light) is also in focus. Figure 2.9 shows an image of solely the kagome structure alongside an image of the kagome structure together with some guided light. The image show that the kagome structure is intact confirming that the cleaving was correctly performed. Furthermore, the mode is clearly guided by the core and exhibits a single lobe, however, the shape is not symmetric with respect to the fiber axis, indicating the excitation of higher order LP_{11} mode alongside the fundamental LP_{01}.

The mode seen in Figure 2.9 right was associated with a transmission efficiency of 63%. By optimizing the alignment using the horizontal microscope the mode shape was optimized, and an efficiency of 80% has been achieved. For this optimization two screw of the mirror mounts were adjusted at the same time as a continuous walk-off, therefore performing an optimization remaining as long as possible in a good injection condition. The resulting mode is shown in Figure 2.10 left. Figure 2.10 right displays another mode obtained with nearly 80% of injection efficiency, where the surrounding kagome structure remains

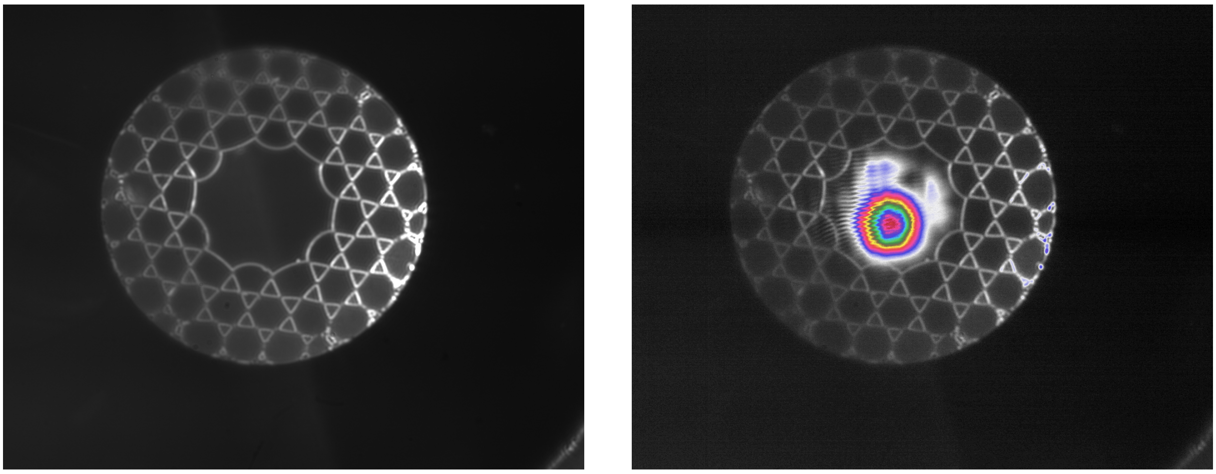


Figure 2.9: *Images taken with a preliminary horizontal microscope setup of the HCPCF kagome structure (left) and of the transmitted mode together with the kagome structure, with a pseudo color map (right). These images were taken with a preliminary setup of the horizontal microscope that provided a more homogeneous illumination, with the red light incident from two direction, making the kagome lattice distinctly visible.*

visible.

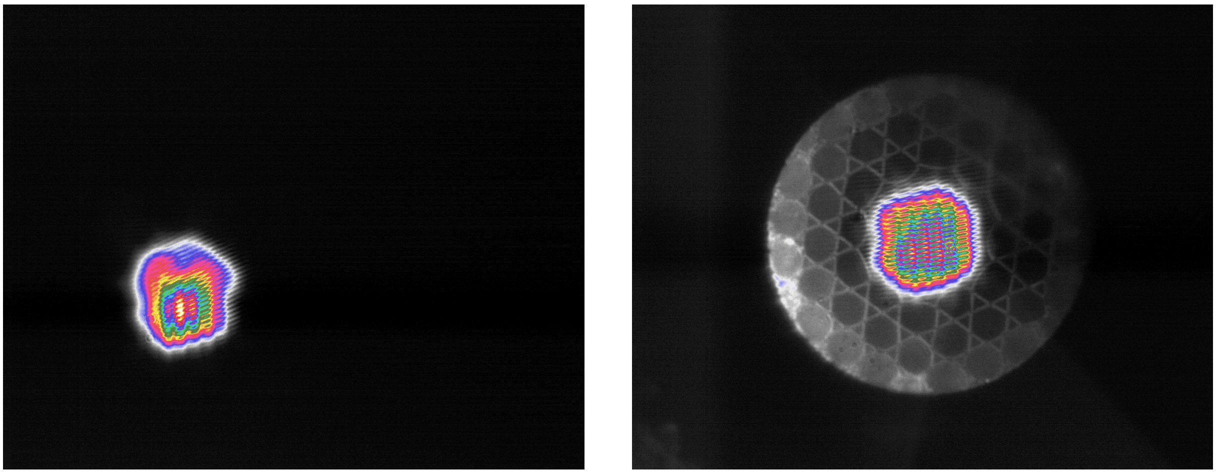


Figure 2.10: *Images captured with the final horizontal microscope setup. A mode with* $\sim$ 80% *of transmission efficiency (left) and a guided mode with the kagome structure visible (right).*

A final optimization was performed without the ocular lens by translating the objective slightly past the working distance and placing the camera approximately 2 m away. With this configuration, the magnification achieved is about $\times 70$ (more than doubling the magnification of the horizontal microscope). At this distance, the light scattered from the kagome structure becomes very faint, complicating the focusing procedure as its signal approaches the noise level. However, employing the FLIR camera, which has a lower read noise and a high sensitivity, the kagome lattice remained resolvable, as shown in Figure 2.11. This figure shows clearly that the near-field mode assumes a hexagonal shape, perfectly fitting the core: as expected the field inherit the symmetry of the medium in which it is confined.

The advantage of this extended projection distance is the significant expansion of the mode, the pixel grid interference patterns disturb the shape far less, allowing finer spatial details to be resolved, further improving the alignment procedure. With this method an efficiency of 94% was achieved. The resulting mode (Figure 2.12 left) is still slightly asymmetric, indicating a residual superposition of higher order modes with the fundamental one.

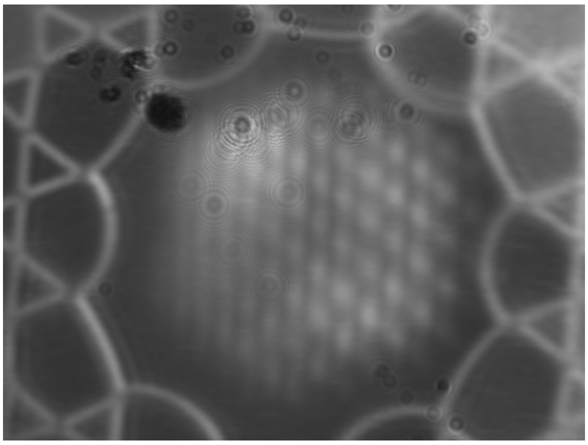


Figure 2.11: *Images obtained with the FLIR camera using the microscope objective without the ocular lens. The mode clearly exhibits a hexagonal shape. Thanks to the FLIR high sensitivity and low read noise, the kagome structure remains visible even in this configuration. Artifacts in the top-left corner are dust grain deposited on the camera sensor.*

Finally, a sweep of laser nominal power was conducted, which changes also the beam waist at the fiber injection. At every tested power the injection alignment was re-optimized, and the best mode profile was obtained at 6.5 W, with an efficiency of 96%. The mode is shown in Figure 2.12 right, in this case the shape is far more symmetric than the previous iterations, which, together with the high transmission efficiency, suggests a significant high share of the fundamental LP_{01} mode.

Qua manca qualcosa sulle due fotcamere, e anche un riferimento alla HCPCF (aggiungerò una sezione)

2.2.3.2 Chamber fiber

After validating the effectiveness of the alignment procedure using the test fiber, the same method was applied to the fiber installed inside the vacuum chamber. Unfortunately, it was not possible to properly illuminate the fiber, and the kagome structure could not be visualized. Using the same 635 nm LED employed for the test fiber, several configurations of focusing lenses and LED positions were tested, however, no light was successfully coupled into the kagome structures. In addition, due to physical constraints imposed by the vacuum chamber geometry, it was not possible to directly illuminate the fiber facet from the outside. One possible limitation might be the presence of a translucent plastic

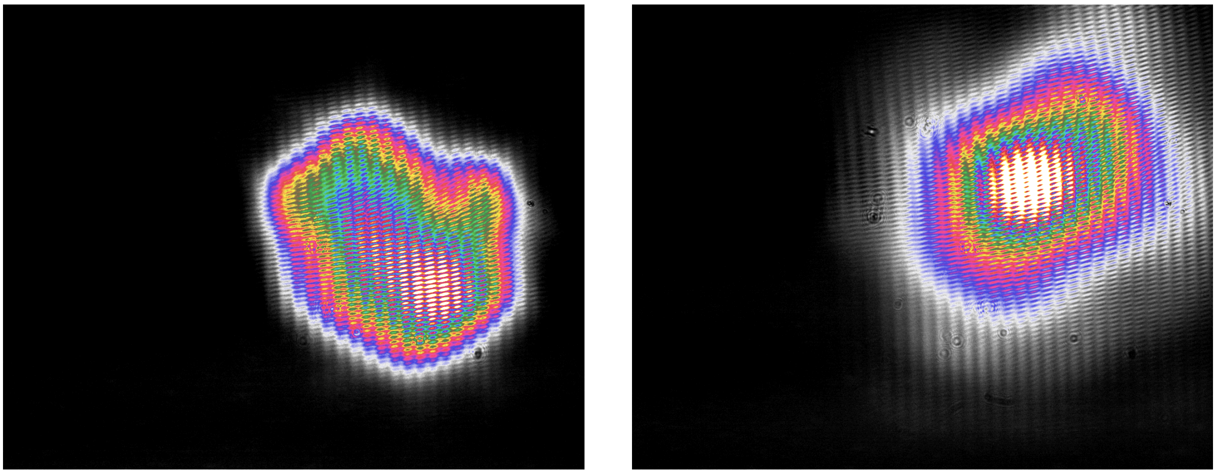


Figure 2.12: *Images obtained with the objective without the ocular lens. Mode with* 94% *of injection efficiency (left). Optimized mode with nominal laser power set to 6.5 W which achieved* 96% *of transmission efficiency (right).*

casing surrounding the fiber. The casing may diffuse the light of the LED preventing it to reach the kagome structure. Unlike the images obtained with the test fiber, in this case the jacket appears illuminated, suggesting that the light propagated in the jacket rather than reaching the fiber kagome structure.

Furthermore, the need to create a virtual image outside the chamber required the use of multiple lenses (see Section 2.2.2.3). While this approach enabled imaging of the fiber facet, it also introduced non-negligible chromatic aberrations. Figures 2.13 left and 2.13 right show respectively a preliminary image of the focused fiber and the same configuration with light guided through the fiber. In the latter case, the optical mode is in focus while the fiber facet appears out of focus.

The mode obtained after the injection alignment procedure is shown in Figure 2.14, which exhibited $\sim 50\%$ of transmission efficiency. It should be noted that this efficiency values was measured after the light propagated through a chamber window, two lenses and two mirrors, all of which introduce transmission and reflection losses. Moreover, the fiber installed in the chamber is longer than the test fiber. Although propagation losses associated with the additional length are should be negligible, the increased length requires several bends in the fiber routing, which are known to induce leakage losses in optical fibers.

The best mode obtained after the optimization procedure is displayed in Figure 2.14. It shows a clear asymmetry, indicating the presence of a superposition of various LP modes. Although, it should also be considered that, in order to enter the vacuum chamber, the fiber must be tightened in a ferrule by a swage-lock fitting. This mechanical stress can deform the fiber microstructure, altering its guiding properties and degrading the mode shape quality.

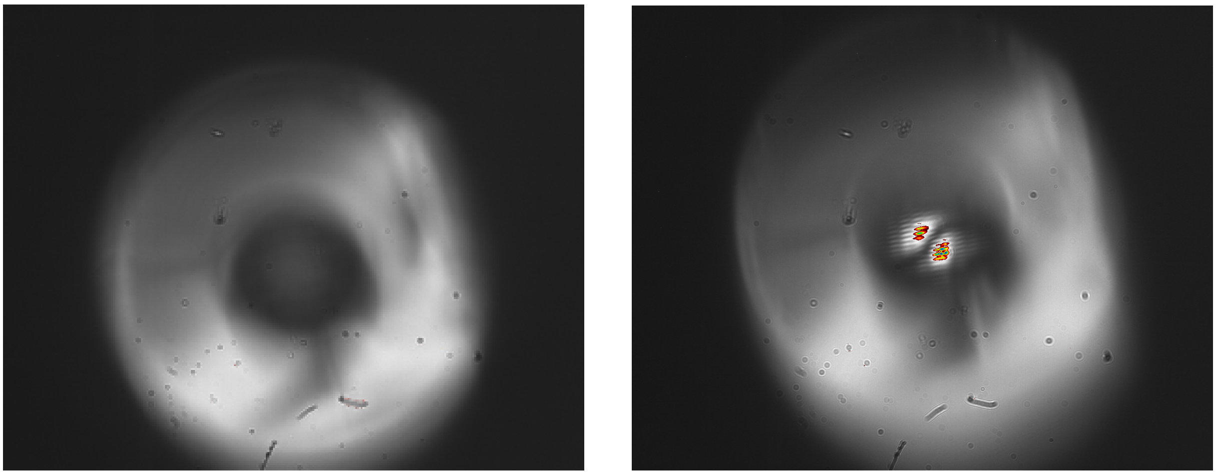


Figure 2.13: *Preliminary images taken with a zoom about* $\times 10$ *of the fiber facet in the vacuum chamber (left) and the fiber with a non-optimized mode (right). In contrast to the test fiber images, here, the kagome structure is not illuminated, whereas the outer jacket is. The mode exhibits a clear* LP_{11} *profile. Note that focusing the mode bring the fiber out of focus, this makes the mode seem non-centered with respect to the fiber core.*

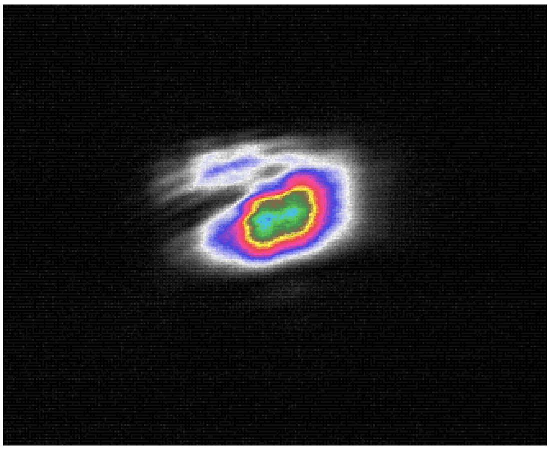


Figure 2.14: *Final mode profile obtained after the injection optimization (magnification about* $\times 10$*). The mode shape is a bit deformed, meaning that it is a superposition of fundamental LP modes. The image is cropped.*

2.3 Experimental Procedures and analysis methodology

The first step of the experimental sequence is the capture of atoms in the MOT, using the optimal parameters previously determined in for this apparatus [10]. The MOT is realized with an axial magnetic field gradient of $B' = 9.32$ G/cm (corresponding to a current of 8 A), a cooler laser detuning of $\delta = -13$ MHz, and all compensation coil currents set to 0 mA. The MOT is loaded for a total duration of 5 s.

After the loading phase, the magnetic field is switched off, initiating the optical molasses phase. In the absence of the magnetic field, the atomic temperature decreases, however the confinement force is lost, and the atomic cloud begins to expand. At the same time, under the influence of gravity, the cloud starts to fall while experiencing a viscous-like damping force due to its interaction with the laser light. The temperature achieved during the molasses phase scales as $I/|\delta|$, where I is the laser intensity and δ is the detuning [11]. However, this relation remains valid as long as $|\delta|$ is sufficiently small to not reduce significantly the optical cycle rate $\Gamma_{\mathrm{p}} \propto 1/\delta^2$. Additionally, the detuning must not be so large to induce a significant excitation probability of the neighboring $F = 2 \rightarrow F' = 2$ transition. Furthermore, optimal molasses performance is obtained when the magnetic field is effectively zero. For this reason 5 ms after the MOT magnetic field is switched off, the shim coil currents are adjusted to compensate for the Earth's magnetic field and any residual magnetic fields present in the experimental environment. The molasses is performed with a detuning different from that used for the MOT phase, it is changed with a ramp of 20 steps with a total duration of 5 ms. The ramp stars 50 μs after the shim coils are changed to their molasses value.

2.3.1 Atom Imaging

Two different CMOS cameras were used to image the atoms inside the vacuum chamber:

- **Teledyne FLIR BFS-U3-04S2M-CS**, monochromatic camera mounting the Sony IMX287 sensor, with a resolution of 720×540, and a pixel size of 6.9 μm $\times$ 6.9 μm.

- **Allied Vision Manta G-419 NIR**, monochromatic camera mounting the CMOSIS/ams CMV4000 CMOS sensor, with a resolution of 2048×2048, and a pixel size of 5.5 μm $\times$ 5.5 μm.

The FLIR camera was primarily used during the optimization of the molasses phase, whereas, the Manta camera was employed for measurements of the atom capture efficiency of the ODT. The Manta was preferred with respect to the FLIR because of its higher sensitivity at 780 nm and its higher pixel density, which helped the imaging of the small region where the ODT is present.

Both cameras were equipped with a Thorlabs MVL50M23 adjustable objective lens with a focal length of 50 mm, and a minimum working distance of 200 mm, corresponding to a minimum magnification of ~ 0.33. The objective was adjusted to focus on the HCPCF inside the vacuum chamber, ensuring that the atomic cloud, located at a similar distance, was also in focus. Both cameras were externally triggered by signals sent by the FPGA control system.

Since the temperature determination of the atomic cloud requires knowledge of its physical dimensions (see Section 2.3.2), the FLIR camera had to be calibrated.

The first calibration method was based on the measure of the diameter in pixel of the fiber inside the vacuum chamber. By turning on the MOT beam, the fiber gets illuminated, and a picture is taken. Using this method the fiber diameter of ~ 316 μm was found to correspond to 8 pixels. This yielded a calibration factor of 39.5 μm/px, corresponding to a magnification of approximately 0.17. However, the measurement uncertainty was significant: assuming an uncertainty of one pixel on each edge of the fiber image, the measured diameter becomes (8 ± 2) px, resulting in a calibration factor of (40 ± 10) μm/px. Consequently, this method provides only a rough estimate of the imaging magnification.

A second and more accurate calibration method was therefore employed. A mirror was positioned in front of the camera without modifying the objective setting, and a ruler with a visible scale was positioned so that it was in focus. This ensured that the ruler was located at the same distance from the objective as the fiber inside the vacuum chamber. With this method 2 cm corresponded to (471 ± 2) px. Since the instrumental uncertainty associated with a single measurement was negligible, the calibration was performed by taking five independent measurement, with the ruler repositioned at focus each time. This procedure yielded a final calibration factor of (43.4 ± 0.3) μm/px, corresponding to a magnification of 0.16.

2.3.2 Temperature measurements

Temperature measurements were performed to optimize the molasses parameters, with the goal of achieving the lowest possible temperature.

The temperature of the atomic sample was determined by measuring its ballistic expansion during free fall. Assuming a non-interacting gas and considering a single spatial coordinate, if an atom is found at x_0 at $t = 0$, after a free expansion time t its position is given by

$$x(t) = x_0 + v_x t. \tag{2.8}$$

Taking into account the distributions of both positions and velocities, and assuming that these quantities are statistically independent, the variance of their sum is equal to the sum of their variances. Therefore,

$$\sigma_x^2(t) = \sigma_{x,0}^2 + \sigma_{v_x}^2 t^2, \tag{2.9}$$

where $\sigma_{x,0}$ is the standard deviation of the position distribution at $t = 0$.
According to classical energy equipartition theorem, the temperature associated with a given spatial direction is related to the squared mean velocity along that direction:

$$\frac{1}{2} m \langle v_x^2 \rangle = \frac{1}{2} k_B T_x, \tag{2.10}$$

where m is the atomic mass and k_B is the Boltzmann constant.
In the frame of reference of the cloud center of mass, the atomic cloud is globally at rest, thus $\langle v_x \rangle = 0$ and

$$\langle v_x^2 \rangle = \sigma_{v_x}^2 = \frac{k_B T_x}{m}. \tag{2.11}$$

Substituting this result into 2.9 one gets:

$$\sigma_x^2(t) = \sigma_{x,0}^2 + \frac{k_B T_x}{m} t^2. \tag{2.12}$$

Using the atomic mass of the ^{87}Rb

$$m = m_A(^{87}R) = 86.9 \text{ u} \approx 1.443 \times 10^{-25} \text{ kg}, \tag{2.13}$$

and acquiring images at different times of flight (TOF), equation (2.12) can be fitted obtaining T_x as a fitting parameter.
The complete acquisition and analysis procedure is summarized below:

1. First a MOT is loaded, followed by the molasses phase. For the molasses different parameters and durations were tested in order to find the best configuration that minimized the temperature (see Section 3.1)

2. After the molasses stage, all cooling beams are switched off, allowing the atomic cloud to expand and fall freely. Using the FLIR camera images are taken at different TOFs, obtaining the intensity distribution $I(TOF)$. For each TOF a corresponding background image $I_{\text{bkg}}(TOF)$ is also acquired. The background image is taken by waiting a sufficiently long time (1s) ensuring that the cloud has completely disappeared and only the background signal remains.

   It should be noted that, due to the time required by the FLIR camera to transfer an image to the computer, it is not possible to take multiple TOF image during the same experimental run, therefore each image requires a new MOT loading and molasses sequence.

3. The background is subtracted

$$I_{\text{clean}}(TOF) = I(TOF) - I_{\text{bkg}}(TOF), \tag{2.14}$$

   and the associated Poisson uncertainty is calculated as

$$\sigma_{\text{N}} = \sqrt{I(TOF) + I_{\text{bkg}}(TOF)}. \tag{2.15}$$

4. A two-dimensional Gaussian is fitted to each image (TRF minimization algorithm) to extract the cloud widths σ_x and σ_y along the two principal axis.

5. Equation (2.12) is fitted independently for both directions, obtaining the temperatures T_x and T_y. Then, the final temperature T is calculated as their average.

2.3.3 ODT loading measurements

In the analysis of the ODT, two main figures of merits are taking into account, namely the loading efficiency η and the intensity integral of the ODT peak, that is directly proportional to the number of captured atoms. They are measured by comparing fluorescence images acquired with the ODT either present or absent during the MOT and molasses stages. Since only a small fraction (order of 1/1000) of the atoms is expected to be captured by the ODT, the analysis relies on differential imaging and careful background normalization to isolate the signal of the trapped atoms from fluctuations in the total atom number. The complete analysis procedure is summarized below.

1. Two identical experimental sequences of MOT and molasses are consecutively performed with the same parameters. In one sequence, the ODT is kept on throughout the entire cycle starting from the MOT loading phase, whereas in the other the ODT remains off. After a TOF of 30 ms, fluorescence images of the atomic distribution denoted by I_{on} and I_{off} are taken with the Manta camera. During the TOF the atom not trapped in the ODT expands enhancing the visibility of those confined in the ODT. Each pair of measurements is repeated five times

2. These images are cropped to a region of interest of 530 × 380 pixels centered on the atomic cloud. This removes the region where the HCPCF is visible as well as peripheral areas containing only background signal.

3. To improve the signal-to-noise ratio, the images corresponding to each experimental condition are averaged:

$$\bar{I}\text{on} = \langle I_{\text{on}} \rangle \qquad \bar{I}\text{off} = \langle I_{\text{off}} \rangle \,.$$

This is important because the number of atom loaded into the ODT is expected to be small.

4. The averaged images are converted into one dimensional density profiles by integrating along the columns. This further increase the signal-to-noise ration.

5. The total number of atoms in the molasses exhibits slow fluctuations even when the experimental parameters are unchanged. These variations are attributed to slow drifts in the chamber pressure and to thermal effects in optical components, such as the AOMs. Since the ODT signal is small, due to the correlated nature of the drifts, despite the average of different images the difference would result in a background different from zero that is comparable to the ODT signal amplitude. To compensate this effect, a rescaling procedure is applied to the background images. This method computes a scale factor, k, which enforces a zero integrated difference outside the ODT region.

   The procedure is implemented as follows:

   i. An exclusion region centered on the ODT position, $x_{\text{ODT}} \pm R_{\text{window}}$ is defined. All remaining positions constitute the outside region.

   ii. A linear background $L_i(x)$ is estimated by interpolating the edges of each intensity profile. After subtracting the background, the integrated signal outside the ODT region is computed for both the ODT-on and ODT-off profiles ($A_{\text{on}}^{\text{out}}$ and $A_{\text{off}}^{\text{out}}$) as

$$A_i^{\text{out}} = \int_{\text{outside}} \Big(P_i(x) - \bar{I}_i(x)\Big)\, dx \qquad i \in \{\text{on, off}\} \tag{2.16}$$

   iii. The normalization factor is given by the ratio of these integrated areas:

$$k = \frac{A_{\text{on}}^{\text{out}}}{A_{\text{off}}^{\text{out}}} \tag{2.17}$$

6. The final rescaled profile, which isolates the signal of the atoms trapped in the ODT, is obtained as:

$$\Delta\bar{I}(x) = \bar{I}_{\text{on}}(x) - k \cdot \bar{I}_{\text{off}}(x) \tag{2.18}$$

7. Two independent fits consisting of a Gaussian function and a linear background are performed. The first fit is applied to the ODT-off profile $\bar{I}_{\text{off}}(x)$,while the second is applied to the differential profile $\Delta\bar{I}(x)$.
   The areas $A_{\text{off}}^{\text{Gaus}}$ and $A_{\text{ODT}}^{\text{Gaus}}$ are calculated from the fit results, and the capture efficiency is calculated as

$$\eta = \frac{A_{\text{ODT}}^{\text{Gaus}}}{k \times A_{\text{off}}^{\text{Gaus}}} \tag{2.19}$$

Chapter 3

Result And Discussion

Following the experimental and analysis procedures explained in Chapter 2, the ODT capture efficiency was characterized and optimized. This chapter presents the corresponding experimental results.
The first stage of the optimization process consist of minimizing the molasses temperature through the adjustment of the cooling laser detuning, the AOM amplitude, and the compensation coil currents.
Subsequently, the ODT loading is optimized by exploiting the compensation coils to displace the MOT after loading. This optimization aims to maximize the spatial overlap between the atomic cloud and the ODT beam, thereby enhancing the capture efficiency.

3.1 Molasses temperature optimization

Before performing any optimization, the six cooling beams were realigned to minimize their overlap mismatch at the MOT position. This procedure is crucial since thermal drifts can induce misalignments in the optical components over time.
Preliminary values for the compensation coil currents were obtained prior to the realignment using the Release-and-Recapture (R&R) technique:

$$\begin{aligned} I_{\mathrm{x}} &= 80\ \mathrm{mA}, \\ I_{\mathrm{y}} &= -200\ \mathrm{mA}, \\ I_{\mathrm{z}} &= -400\ \mathrm{mA}. \end{aligned} \tag{3.1}$$

These values were used as the starting point for the subsequent optimization procedure.

3.1.1 Detuning frequency optimization

The first optimization step consisted of measuring the molasses temperature for different values of the cooling laser detuning. The detuning was varied from -24.4 MHz ($4.0\ \Gamma/2\pi$)

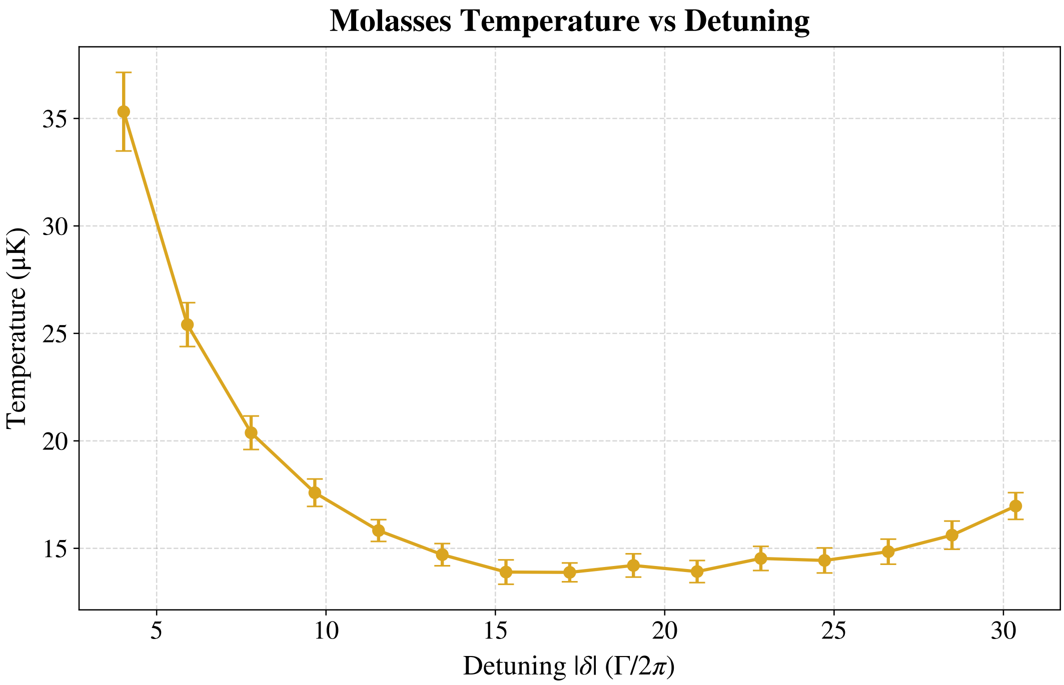


Figure 3.1: *Optical molasses temperature as a function of laser detuning. The horizontal axis represents the absolute value of the applied red detuning* $|\delta|$*, normalized to the natural linewidth* $\Gamma/2\pi$*. The data exhibits optimal sub-Doppler cooling in the region* $|\delta| \approx 15-20$*, reaching a minimum temperature of* 13.9 ± 0.4 μK*. For detunings up to* $|\delta| \approx 15$*, the temperature follows the* $1/|\delta|$ *dependence expected for an ideal Sisyphus cooling. At larger detunings, the curve flattens and eventually begins to rise.*

to -184 MHz ($30.4\ \Gamma/2\pi$), with a step size of -11.4 MHz.

The resulting average temperature is shown in Figure 3.1 as a function of the detuning expressed in units of $\Gamma/2\pi$. At small detunings, the data follow the expected $1/|\delta|$ dependence typical of the Sisyphus cooling. As the detuning increases, two phenomena become relevant: a significant drop in the optical scattering rate $\Gamma_p \propto 1/\delta^2$ and the approach of the transition $F = 2 \rightarrow F' = 2$ located 267 MHz below the cooling transition $F = 2 \rightarrow F' = 3$ (see Figure ref??). These effects cause the temperature curve to flatten in the region $|\delta| \approx 15 - 20$ and eventually to rise again for larger detunings.

The lowest average temperature was obtained for a normalized detuning of $|\delta| = 17.2$ corresponding to an absolute detuning of $|\delta| = 104.2$ MHz. Figure 3.2 shows the fit of equation (2.9) for this optimal configuration. The resulting temperatures along the Cartesian axes are $T_x = (14.1 \pm 0.5)\ \mu$K and $T_y = (13.5 \pm 0.7)\ \mu$K. Their weighted average yields a final temperature of $T = (13.9 \pm 0.4)\ \mu$K.

As observed in Figure 3.2, the data points at long TOF diverge appreciably from the linear trend, consistently falling below the expected values. This behavior is systematically observed across all measurements. The hypothesis is that at sufficiently long TOF the cloud radius becomes non-negligible compared to the probe beam radius, violating the assumption of uniform illumination. Consequently, the outer region of the cloud interacts

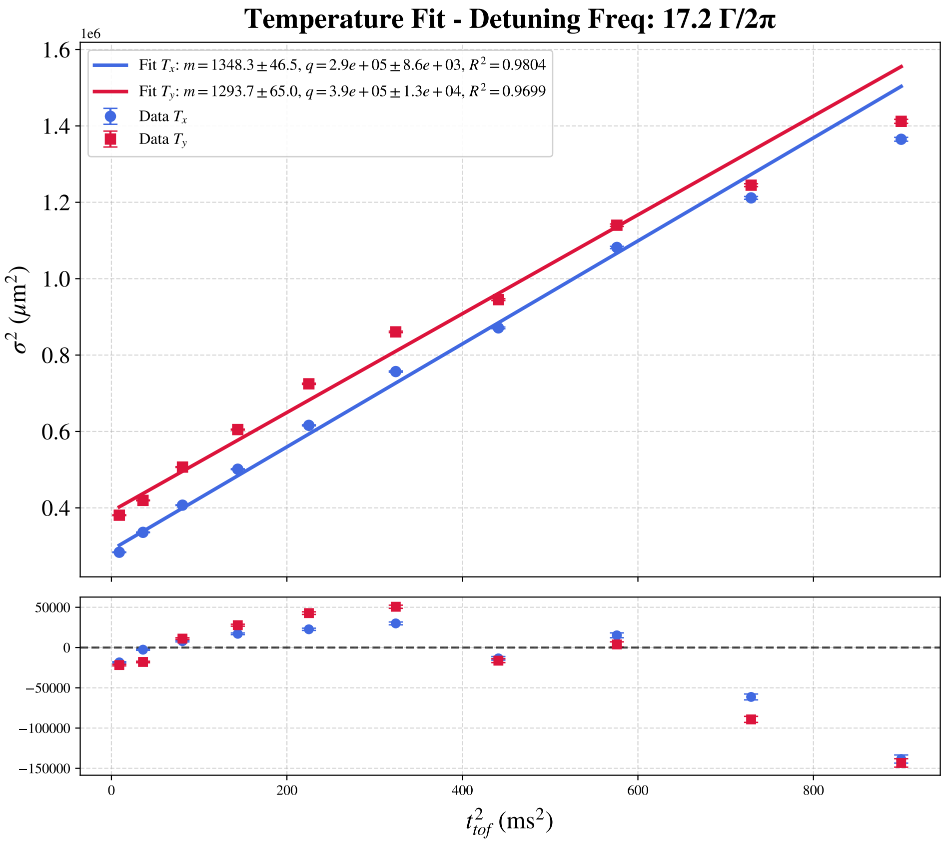


Figure 3.2: *Optical molasses temperature fit for $|\delta| = 17.2\ \Gamma/2\pi$ with residuals. The large residuals of the last three data points from the linear fit are attributed to the cloud expansion, which makes its radial size non-negligible compared to the probe beam profile.*

with a lower intensity field, causing the Gaussian fit to systematically underestimate the true variance. Fortunately, this effect is expected to become less prominent as the temperature is lowered.

3.1.2 AOM amplitude optimization

As a second optimization step, the influence of the cooling beam power was investigated by varying the AOM drive amplitude and the duration of its transition ramp, which is applied at the beginning of the molasses phase. The AOM amplitude, expressed in arbitrary units (a.u.), controls the RF amplitude driving the AOM. Three values were tested: 385, 475 and 550 a.u., corresponding respectively to approximately 0.7, 0.85 and 1.0 times the AOM amplitude during the MOT phase. Additionally, ramp durations of 5 ms and 20 ms were tested for the transition from the MOT value to the molasses value. These measurements were repeated for different values of the cooling detuning. The optimization results are shown in Figure 3.3. Although small variations in the expected values of the temperature can be observed for different values of the AOM amplitude, these differences fall within the experimental error bars. Therefore, no statistically significant dependence of the molasses

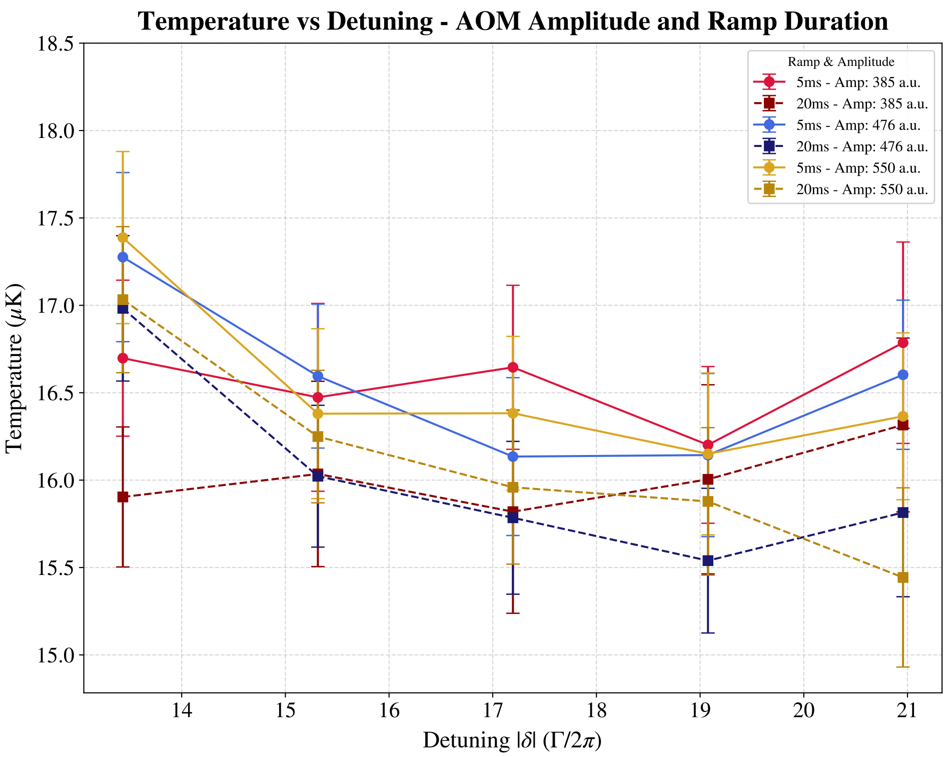


Figure 3.3: *Optical molasses temperature versus cooling detuning for different AOM amplitude and ramp durations. No clear optimal AOM amplitude is observed as the variations remain within the uncertainty. Temperatures corresponding to a ramp duration of 20 ms (dashed lines) are systematically lower than those obtained with a 5 ms ramp (solid lines).*

temperature on the AOM amplitude was found, and no clear optimum could be identified within the tested range. Lower amplitudes were not investigated to avoid operating the AOM too far from its optimal efficiency regime.

Conversely, as shown in Figure 3.3, the temperatures obtained with a 20 ms ramp are systematically lower than those obtained with a 5 ms ramp, yielding an average temperature reduction of approximately 0.5 μK. This improvement was relatively small, and was considered sufficient for the scope of the experiment. Since further optimization on the order of a few tenths of a microkelvin was not required, no other ramp times were tested. Consequently, a ramp duration of 20 ms was selected for the subsequent measurements.

It is worth noting that during the ODT loading characterization, this optimization was found to introduce additional fluctuations into the system. For this reason, the subsequent optimization of the compensation coil currents was carried out using an AOM amplitude of 385 a.u. (corresponding to the lowest temperature measured at $|\delta| = 17.2\ \Gamma/2\pi$) with a ramp duration of 20 ms. However, these parameters were later readjusted during the ODT optimization. Nevertheless, the measurements presented in this section revealed that their influence on the molasses temperature is limited to a few tenths of a microkelvin, and therefore does not significantly affect the final achieved temperature.

3.1.3 Compensation field optimization

The final stage of the molasses optimization procedure consisted of adjusting the compensation magnetic field. The compensation coil currents are set to the molasses values 5 ms after the MOT magnetic field had been switched off.

The optimization was performed sequentially. First, the z-axis current was optimized while keeping the x- and y-axes currents fixed to their preliminary values (3.1). Then the x-axis current was optimized using the newly found optimal z-value. Finally, the y-axis current was optimized using the optimal currents for both the z- and x-axes. The results are shown in Figures 3.4, 3.5 and 3.6 for the z-, x-, and y-axis coils respectively.

The resulting optimized currents, used in the subsequent ODT loading stage, are:

$$\begin{aligned} I_{\mathrm{x}} &= -15 \text{ mA}, \\ I_{\mathrm{y}} &= -185 \text{ mA}, \\ I_{\mathrm{z}} &= -230 \text{ mA}. \end{aligned} \tag{3.2}$$

The z-axis compensation coil current deviated the most from its preliminary value, while the x-axis current underwent a smaller but still appreciable correction. In contrast, the y-axis coil current remained close to its initial value.

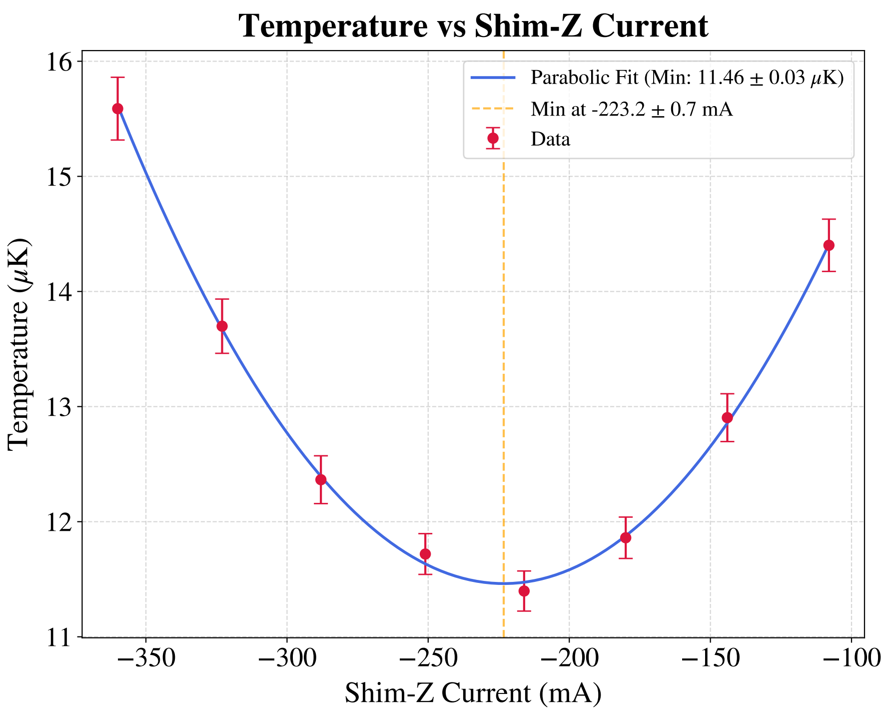


Figure 3.4: *Optical molasses temperature versus Shim-Z current. The parabolic fit yields a minimum temperature* $T = (11.46 \pm 0.03)$ μK *at a current* $I_z = (-223.2 \pm 0.7)$ *mA.*

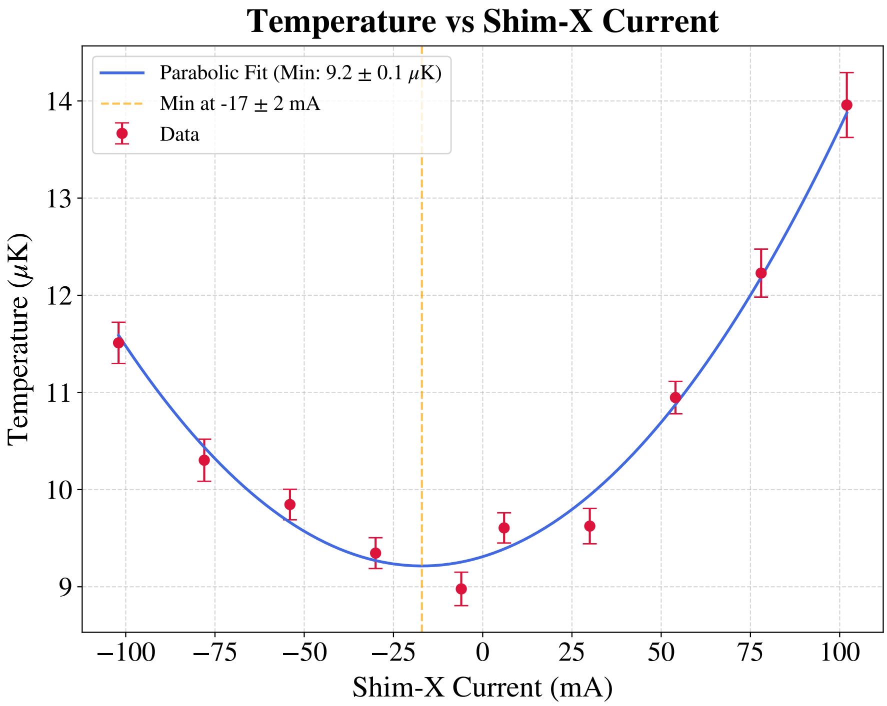


Figure 3.5: *Optical molasses temperature versus Shim-X current. The parabolic fit yields a minimum temperature* $T = (9.2 \pm 0.1)$ μK *at a current* $I_x = (-17 \pm 2)$ *mA.*

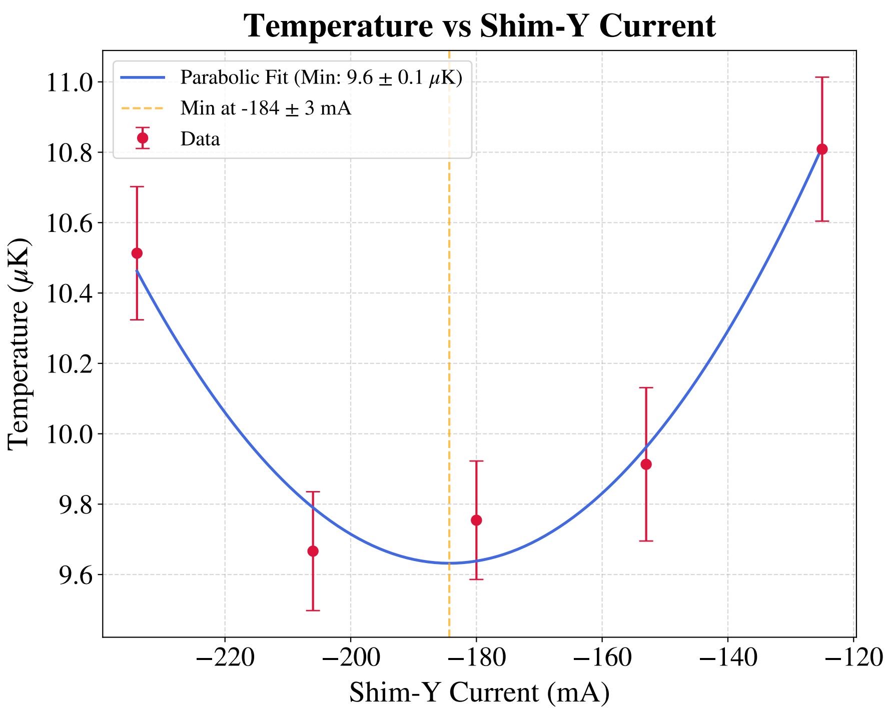


Figure 3.6: *Optical molasses temperature versus Shim-Y current. The parabolic fit yields a minimum temperature of* $T = (9.6 \pm 0.1)$ μK *at a current* $I_y = (-184 \pm 3)$ *mA.*

3.2 ODT loading optimization

The optimal parameters obtained for the molasses were used for the optimization of the ODT loading. This procedure primarily focused on tuning the compensation magnetic field. The underlying concept is to first load the MOT and then, prior to the molasses phase, exploit the compensation coils to spatially shift the MOT position.
Because the MOT forms at the zero of the magnetic quadrupole field, adjusting the compensation coil currents effectively translates the field zero, displacing the trapped atomic cloud. Moving the MOT away from the optimal intersection volume of the beams inevitably causes some atom loss and a temperature increase. However, this is a necessary trade-off to achieve a better spatial overlap with the ODT beam, which ultimately maximizes the number of atoms transferred into the dipole trap. As long as this spatial shift is small, the optimal molasses parameters derived in the previous sections remain effective at nullifying the residual magnetic field, requiring no further adjustments
Specifically, after the initial MOT loading phase, the compensation coil currents are ramped to their ODT loading values over a duration of 200 ms, discretized in 20 steps. This smooth translation is crucial to minimize atom loss and heating during the transport process.
Similarly to the molasses phase, preliminary current values for the ODT loading, obtained prior to the optical realignment and the molasses optimization, were available as a starting point:

$$\begin{aligned} I_{\mathrm{x}} &= -200 \text{ mA} \\ I_{\mathrm{y}} &= -350 \text{ mA} \\ I_{\mathrm{z}} &= 50 \text{ mA}. \end{aligned} \tag{3.3}$$

3.2.1 Shifted MOT

3.2.1.1 Preliminary measurements

Initial measurements were performed using the FLIR camera, employing the optimal parameters derived from the molasses temperature optimization and the preliminary compensation coil currents for the ODT loading. The data were analyzed to characterize the ODT capture efficiency using a procedure analogous to that described in Section 2.3.3, though without background rescaling. The resulting intensity difference images are shown in Figure 3.7 (2D image) and Figure 3.8 (1D integrated profile). Visual inspection of the 2D image does not reveal any distinct brighter region attributable to the atoms captured by the ODT. Conversely, the 1D integrated profile shows a localized intensity increase spanning ~ 4 pixels around the fiber position. Subsequent measurements revealed a similar behavior, confirming that this localized peak corresponds to the atoms trapped in the

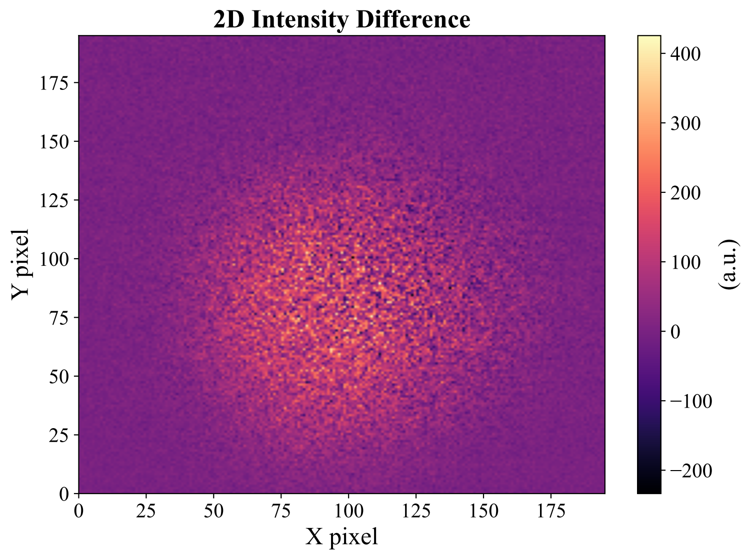


Figure 3.7: *Intensity difference 2D image, obtained with the FLIR camera and a 20 ms AOM amplitude ramp duration during the molasses phase. The intensity distribution does not show any brighter area that can be attributed to the presence of the ODT.*

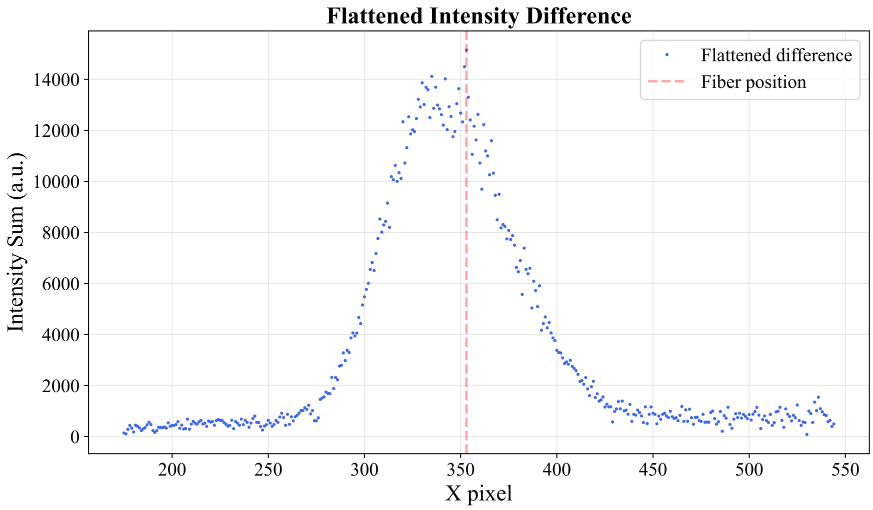


Figure 3.8: *Intensity difference 1D integrated profile, obtained with the FLIR camera and a 20 ms AOM amplitude ramp during the molasses phase. The red vertical line indicates the approximate position of the fiber center. A few higher-intensity pixels are recognizable corresponding to the ODT position, but the averaging and subtracting procedures are not effective in removing the background.*

ODT. However, the signal amplitude was often too small to be reliably distinguished from the noise. Since the 1D integration is explicitly intended to enhance the signal-to-noise ratio (SNR), the persistence of a poor SNR highlighted a critical issue. Furthermore, the presence of a residual background pedestal significantly larger than the ODT signal indicates underlying oscillatory noise in the system that is not averaged by the acquisition of five images, suggesting either a large standard deviation or a correlated shot-to-shot drift. Beyond the noise, the magnitude of the ODT signal itself was a primary concern. Even

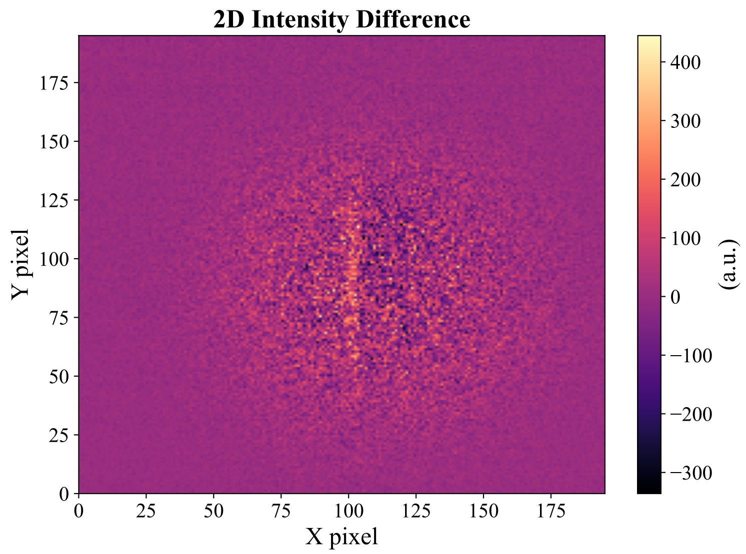


Figure 3.9: *Intensity difference 2D image, obtained with the FLIR camera and a 5 ms AOM amplitude ramp duration during the molasses phase. A brighter area corresponding to the ODT is clearly visible.*

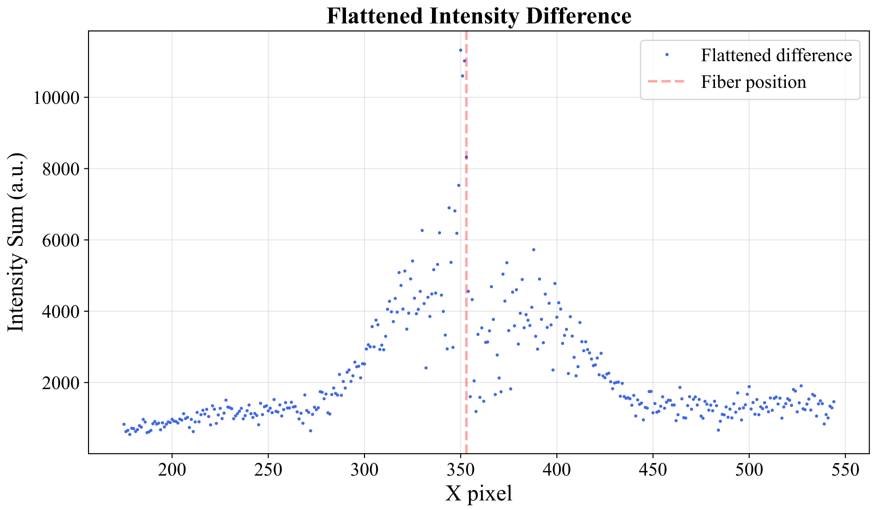


Figure 3.10: *Intensity difference 1D integrated profile, obtained with the FLIR camera and a 5 ms AOM amplitude ramp during the molasses phase. The red vertical line indicates the approximate position of the fiber center. With the 5 ms ramp, the averaging and the subtraction become more effective in separating the ODT signal from the background. However, the problem of the small number of pixels representing the ODT signal is still present.*

far from an optimal atomic-cloud-ODT overlap, a higher initial capture was expected. Additionally, the limited spatial extent of the ODT signal (occupying only a few pixels) posed a severe analytical challenge. Such poor spatial resolution precludes reliable 2D Gaussian fitting, and extracting the signal area via direct pixel summation introduces systematic errors. In summary, the diagnostic capabilities were limited by two main factors: first, the residual background in the subtracted images overwhelmed the weak ODT

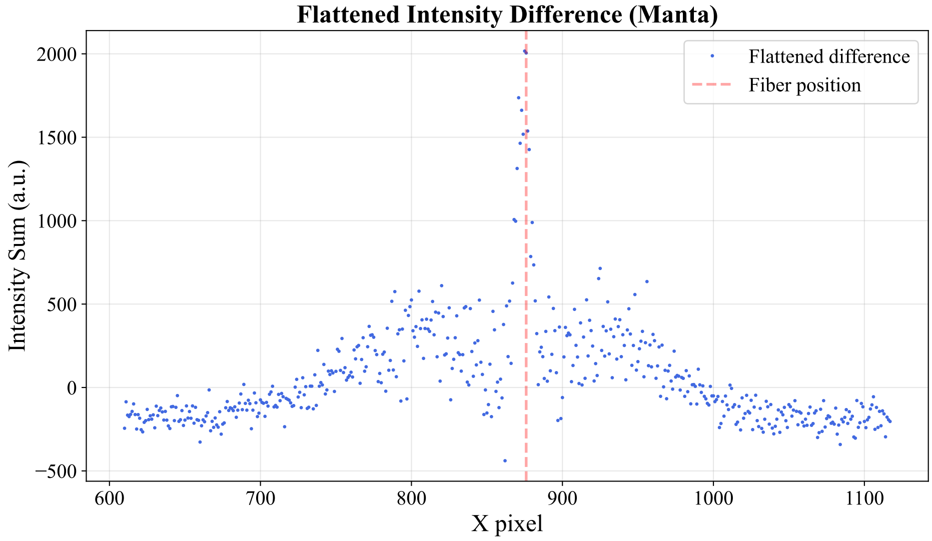


Figure 3.11: *Intensity difference 1D integrated profile, obtained with the Manta camera and a 5 ms AOM amplitude ramp during the molasses phase. The combination of the Manta's smaller pixel size and the closer mounting position effectively increased the spatial resolution of the ODT signal.*

signal; second, the small waist of the ODT beam resulted in an inadequate number of illuminated pixels for robust analysis.

To address the poor SNR, the molasses parameters were systematically investigated. It was found that the noise was strongly dependent on the AOM amplitude ramp during the molasses phase. In particular, reducing the ramp duration from 20 ms to 5 ms significantly improved the background subtraction and increased the SNR. Furthermore, to minimize intensity fluctuations induced by the AOM driver, the target amplitude was set to 480 a.u. This value corresponds to the saturation regime of the AOM diffraction efficiency, which acts as a working point with a minimum derivative ($dP/dV \approx 0$). Operating in this saturation region intrinsically suppresses amplitude noise, thereby stabilizing the actual optical power delivered during the molasses phase.

The results obtained with this optimized configuration are shown in Figures 3.9 (2D difference) and 3.10 (1D integrated profile). In the 2D plot, a distinct brighter strip representing the atoms captured in the ODT is now recognizable. By comparing the 1D integrated profiles in Figures 3.8 and 3.10, it is evident that the pedestal in the 5 ms ramp case is suppressed, showing a maximum amplitude and area approximately half those of the 20 ms ramp case. This reduction is sufficient to make the ODT signal emerge clearly from the residual pedestal and noise.

Despite the improved SNR, a critical issue remained: the insufficient spatial resolution provided by the FLIR camera setup. To overcome this limitation, subsequent measurements were performed using the Manta camera, which features a smaller pixel pitch. Additionally, the Manta camera was mounted closer to the experimental chamber to exploit the minimum working distance of the MVL50M23 objective, maximizing the optical magnification. Comparing the apparent fiber width measured by both cameras, it was deduced that the object size in pixels increased by a factor of ~ 2.25 with the Manta setup. This scaling factor arises from the combined contribution of the higher optical magnification and the smaller pixel size, finally yielding a sufficiently resolved ODT signal for robust analysis, as shown in Figure 3.11.

3.2.1.2 Results

The systematic optimization of the compensation coil currents was carried out following the complete analysis procedure detailed in Section 2.3.3, utilizing the Manta camera. The procedure was performed sequentially along the spatial axes in the order $x \to z \to y$. During each scan, the currents of the remaining non-optimized coils were maintained at the preliminary values established in Equation (3.3).

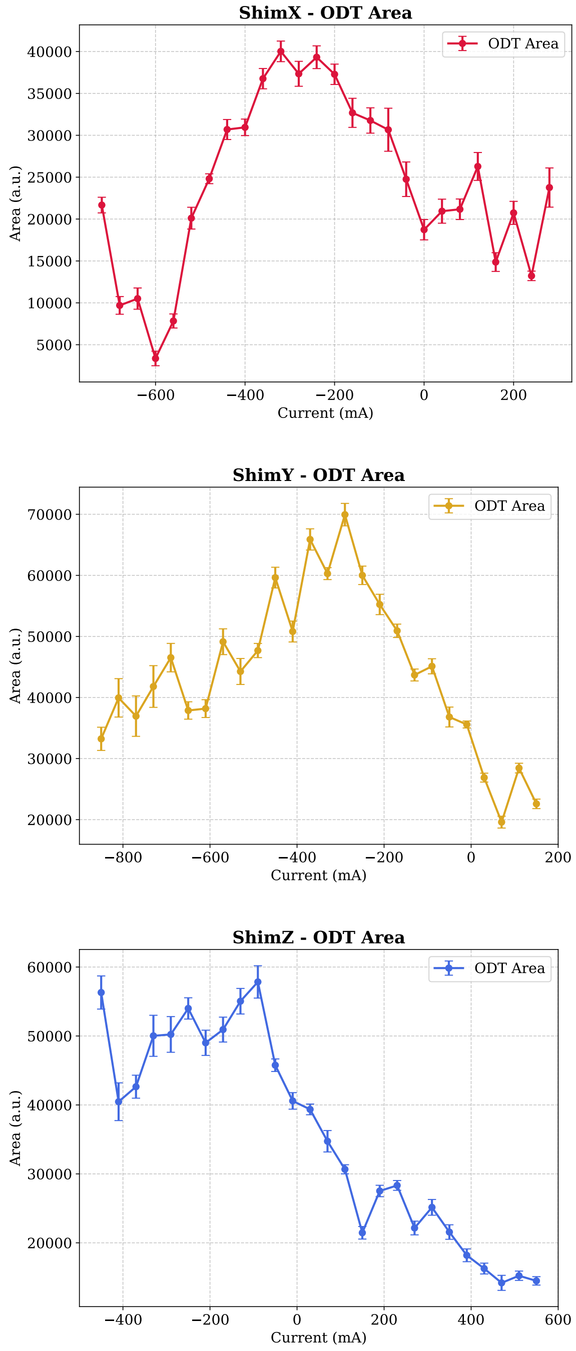
The capture efficiencies extracted from these scans are summarized in Figure 3.12b for the x-, y-, and z-coils. Similarly, Figure 3.12a displays the evaluated ODT signal area for each scan. This area is directly proportional to the total number of captured atoms, making it the physical quantity of fundamental interest. However, since the camera was not calibrated for absolute atom number measurements, this quantity can only be expressed in arbitrary units. Nonetheless, these values remain perfectly comparable provided the camera gain is kept constant. As evident from the plots, the behavior of the capture efficiency mirrors the trend of the ODT area.

This initial optimization yielded the following optimal currents:

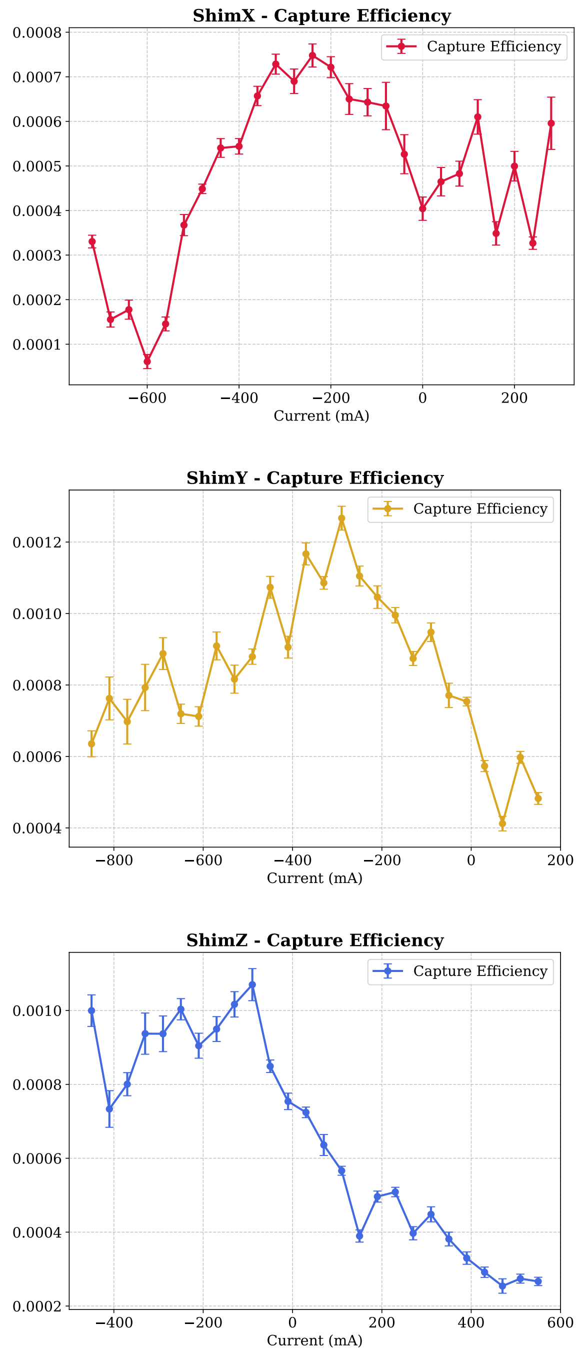
$$\begin{aligned} I_{\mathrm{x}} &= -220 \text{ mA}, \\ I_{\mathrm{y}} &= -290 \text{ mA}, \\ I_{\mathrm{z}} &= -90 \text{ mA}. \end{aligned} \tag{3.4}$$

To confirm the optimization achieved in the initial measurements, a second run was performed, yielding unconventional results regarding the z-axis compensation coil (Figure 3.13). As clearly shown in Figure 3.13a, the signal does not follow an expected parabolic or bell-shaped profile with a clear maximum. Instead, the ODT area and the capture efficiency exhibit an oscillatory behavior. Starting from a current of -1000 mA, the signal increases to a first local maximum at -800 mA, drops until ~ -700 mA, and then rises again to reach subsequent local maxima at ~ -600 mA and ~ -400 mA.

A similar periodic trend is observed in the MOT background area which represents the atomic cloud area measured at the end of the TOF with the ODT turned off (Figure 3.13b). This area exhibits peaks at ~ -800 mA, ~ -600 mA, and ~ -400 mA, matching the ODT area oscillation with good precision. While a proportional increase in the captured ODT fraction is naturally expected when the initial MOT atom number is higher, the

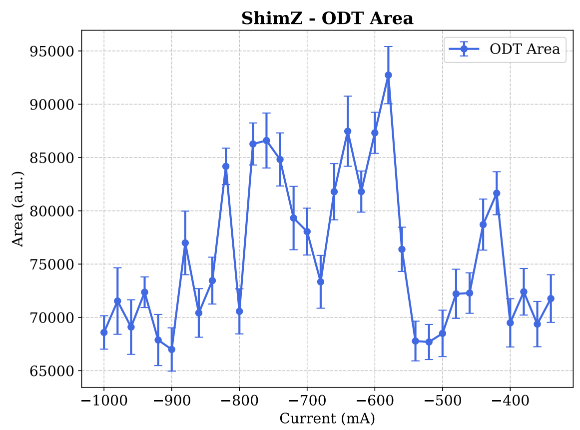


(a) *ODT area for x-, y-, and z-axis scans.*

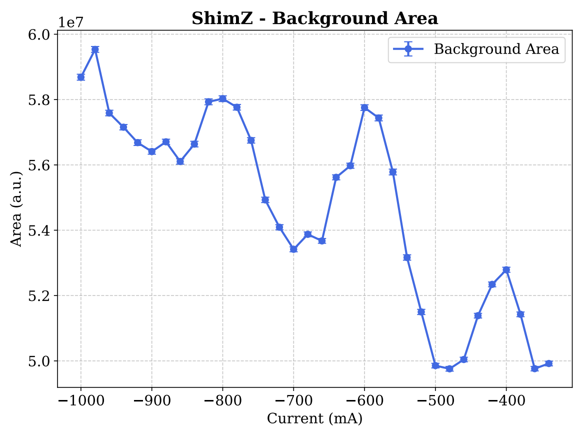


(b) *Capture efficiency for x-, y-, and z-axis scans.*

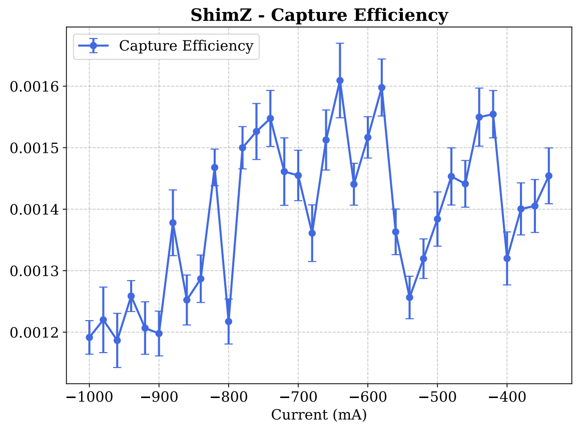
Figure 3.12: *ODT area (left) and capture efficiency (right) for the compensation coil current scans along x-, y-, and z-axes (performed in the order $x \to z \to y$). The qualitative agreement between the two observables is clearly visible for all the investigated axes.*



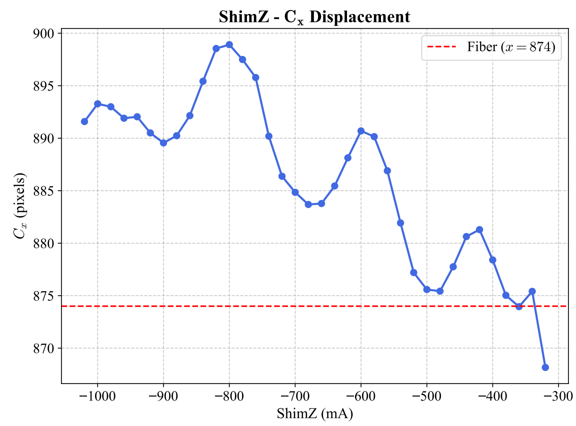
(a)



(b)



(c)



(d)

Figure 3.13: *Optimization results for the second set of measurements of the z-compensation coil current: (a) ODT signal area, (b) background area, (c) capture efficiency, (d) atomic cloud center position projected on the camera's x-axis.*

oscillatory nature of this trend is peculiar. Tracking the x-centroid of the atomic cloud on the camera's image plane (Figure 3.13d) reveals an analogous spatial oscillation, a phenomenon not observed during the scans of the other two compensation axes. The likely physical explanation for this behavior lies in an imperfect spatial overlap of the cooling beams. As the z-axis compensation field is scanned, the position of the magnetic zero shifts along the laboratory z-axis as expected. However, a non-ideal beam intersection generates a complex 3D optical intensity landscape with local gradients. Since the MOT equilibrium position is determined by the local balance of the radiation pressures, the atomic cloud dynamically adapts its position to the local intensity imbalances. Instead of a linear translation along z, the cloud is pushed along a trajectory that locally nullifies the net optical force, resulting in a combined motion in the laboratory z–x plane that projects as a spatial oscillation onto the 2D camera sensor. Furthermore, as the MOT position change in this complex optical landscape, it encounters regions of varying laser intensity. As a consequence, both the cooling efficiency and the trap depth fluctuate accordingly.

Consequently, the atom loss and the final MOT temperature vary depending on the exact spatial position of the cloud. This spatial dependence explains the observed fluctuations in the background are, and directly impacts the subsequent capture efficiency into the ODT.
Finally, the optimal configuration maximizing the ODT loading was found to be:

$$\begin{aligned} I_{\mathrm{x}} &= -220 \text{ mA}, \\ I_{\mathrm{y}} &= -280 \text{ mA}, \\ I_{\mathrm{z}} &= -600 \text{ mA}. \end{aligned} \tag{3.5}$$

This configuration yielded the largest ODT trap area, representing the maximum number of captured atoms, and is associated with a capture efficiency of $(0.160 \pm 0.005)\%$. The corresponding 2D difference image is shown in Figure 3.14.

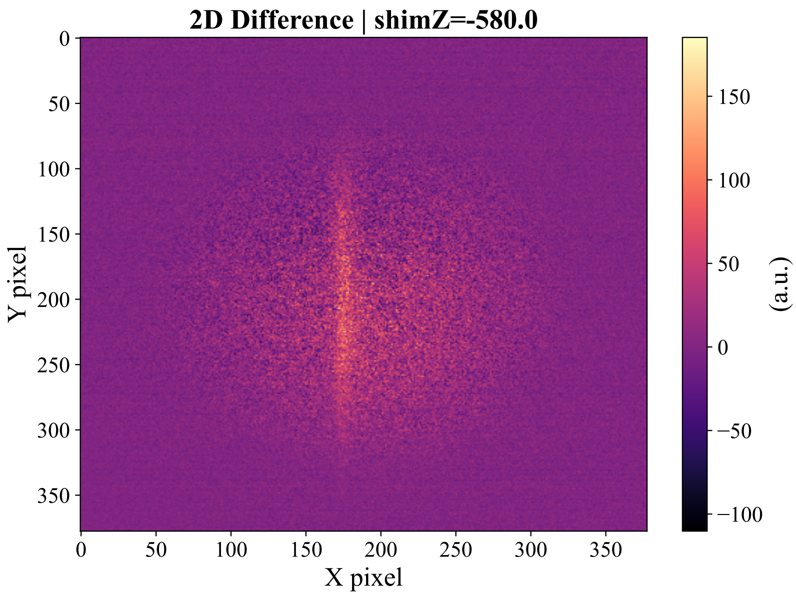


Figure 3.14: *Intensity difference 2D image obtained with the optimal currents:* $I_{\mathrm{x}} = -220$ mA*,* $I_{\mathrm{y}} = -280$ mA*, and* $I_{\mathrm{z}} = -600$ mA*, corresponding to a capture efficiency of* $\sim 0.16\%$*.*

Bibliography

[1] A. Yariv and P. Yeh. *Photonics: Optical Electronics in Modern Communications*. Oxford series in electrical and computer engineering. Oxford University Press, 2007. ISBN: 9780195179460. URL: https://books.google.it/books?id=B2xwQgAACAAJ.

[2] D. Gloge. "Weakly Guiding Fibers". In: *Appl. Opt.* 10.10 (Oct. 1971), pp. 2252–2258. DOI: 10.1364/AO.10.002252. URL: https://opg.optica.org/ao/abstract.cfm?URI=ao-10-10-2252.

[3] Bahaa E. A. Saleh and Malvin Carl Teich. *Fundamentals of Photonics*. 3rd. Hoboken, NJ: John Wiley & Sons, 2019. ISBN: 978-1-119-50687-4.

[4] L. Allen et al. "Orbital angular momentum of light and the transformation of Laguerre-Gaussian laser modes". In: *Phys. Rev. A* 45 (11 June 1992), pp. 8185–8189. DOI: 10.1103/PhysRevA.45.8185. URL: https://link.aps.org/doi/10.1103/PhysRevA.45.8185.

[5] Rudolf Grimm, Matthias Weidemüller, and Yurii B. Ovchinnikov. *Optical dipole traps for neutral atoms*. 1999. arXiv: physics/9902072 [physics.atom-ph]. URL: https://arxiv.org/abs/physics/9902072.

[6] Daniel A. Steck. *Quantum and Atom Optics*. Revision 0.16.8, accessed: 2026-06-01. 2026. URL: http://steck.us/teaching.

[7] Harold J. Metcalf and Peter Straten. *Laser Cooling and Trapping*. Graduate Texts in Contemporary Physics. Springer New York, NY, 1999. DOI: https://doi.org/10.1007/978-1-4612-1470-0.

[8] Claude Cohen-Tannoudji. "Atomic Motion in Laser Light". In: *Fundamental Systems in Quantum Optics*. Ed. by J. Dalibard, J. M. Raimond, and J. Zinn-Justin. Vol. LIII. Les Houches Summer School. Elsevier Science Publishers, 1992.

[9] A. E. Siegman. "How to (Maybe) Measure Laser Beam Quality". In: *DPSS (Diode Pumped Solid State) Lasers: Applications and Issues*. Optica Publishing Group, 1998, MQ1. DOI: 10.1364/DLAI.1998.MQ1. URL: https://opg.optica.org/abstract.cfm?URI=DLAI-1998-MQ1.

[10] Letizia Loparco. "Characterization of a magneto-optical trap of 87 Rb atoms". Supervisor: Prof. Francesco Minardi. MA thesis. University of Bologna, Italy, 2025. URL: https://amslaurea.unibo.it/id/eprint/35347.

[11] J. Dalibard and C. Cohen-Tannoudji. "Laser cooling below the Doppler limit by polarization gradients: simple theoretical models". In: *J. Opt. Soc. Am. B* 6.11 (Nov. 1989), pp. 2023–2045. DOI: 10.1364/JOSAB.6.002023. URL: https://opg.optica.org/josab/abstract.cfm?URI=josab-6-11-2023.